

Research paper

Genetic algorithm-based optimization of combined supercritical CO₂ power and flash-tank enhanced transcritical CO₂ refrigeration cycle for shipboard waste heat recuperation

Fairooz Nanzeeba, Tajwar A. Baigh, Afrida Kabir, Yasin Khan, M. Monjurul Ehsan^{*}

Department of Mechanical and Production Engineering (MPE), Islamic University of Technology (IUT), Board Bazar, Gazipur 1704, Bangladesh

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ABSTRACT

The shipping industry significantly contributes to global trade and economy, but is also responsible for approximately ~3 % of global greenhouse gas (GHG) emissions. Harnessing waste heat from marine vessels to perform useful work is a potential solution to reduce these emissions. Although the Organic Rankine Cycle (ORC) and CO₂ cycles have been studied separately for waste heat recovery, a combined power and refrigeration systems, utilizing both cycles have not been reported. Herein, a combined system, “Marine Energy Recovery System (MERS)”, that integrates the supercritical CO₂ Brayton power cycle with a flash-tank-enhanced transcritical CO₂ refrigeration cycle, is proposed, aiming to generate power and provide cooling simultaneously. The power cycle incorporates an ORC, and a flash-tank is introduced within the refrigeration cycle to improve system performance. Both cycles share a low-temperature recuperator and a gas cooler to further utilize the heat between them. The MERS is evaluated from both a 1st and 2nd law perspective against various performance metrics under different operating conditions, such as: gas cooler pressure, evaporation temperature, and turbine inlet temperature and pressure. Results elucidated that the MERS achieved energy and exergy efficiencies of 54.62 % and 54.90 %, respectively, representing significant improvements over similar systems. The net work output improved by 717.55 kW and the net cooling capacity by 515.7 kW relative to the reference system. Furthermore, a COP of 2.99 outlines its potential as a cooling system. To further enhance MERS's performance, a multi-objective optimization approach is employed, utilizing Artificial Neural Network (ANN) and Genetic Algorithm (GA). Optimal operational conditions yielded an energy efficiency of 74.95 % while maintaining a net work output of 6890.55 kW with gas cooler pressure, turbine inlet temperature, and gas cooler outlet temperature being 9.5 MPa, 461.76 °C and 50 °C respectively. These findings support the reliability and improved design of the MERS for future marine applications.

1. Introduction

The introduction of sustainable and efficient energy systems is receiving a lot of attention due to the world's growing energy consumption and the pressing need to cut greenhouse gas emissions. Traditional energy generation methods are often associated with substantial waste heat, representing a critical area for efficiency improvements and energy recovery. This is especially pertinent in diesel engine-dominated industries such as the shipping industry, which accounts for almost 3 % of the global greenhouse gas (GHG) emissions (Mallouppas

and Yfantis, 2021). Diesel engines are reported to utilize roughly 50 % of the heat energy from the fuel, with the remaining 50 % being lost as waste heat (Larsen et al., 2014). Depending on the mobility of the ship and their placement, most maritime vessels have shipboard power plants that typically uses gas turbines, diesel engines or steam turbines, to provide the necessary electricity for operations. By integrating waste heat recovery (WHR) systems in these engines, the waste heat from these naval components can be conserved, thereby promoting sustainability and economic viability (Lion et al., 2020). Consequently, the thermodynamic WHR systems driving a marine vessel needs to be highly efficient, reliable, and capable of operating under the challenging

Abbreviation: COP, Coefficient of performance; HPC, High-pressure compressor; HRHE, Heat-recovering heat exchanger; HTR, High-temperature recuperator; GC, Gas Cooler; LPC, Low pressure compressor; LTR, Low pressure recuperator; ORC, Organic Rankine cycle; S-CO₂, Supercritical carbon dioxide; T-CO₂, Transcritical carbon dioxide.

^{*} Corresponding author.

E-mail address: ehsan@iut-dhaka.edu (M.M. Ehsan).

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Nomenclature			
Symbols		<i>flash</i>	Flash-tank
<i>T</i>	Temperature (°C)	<i>mixc</i>	Mixing chamber
<i>P</i>	Pressure (MPa)	<i>HX</i>	Heat exchanger
<i>h</i>	Enthalpy (kJ/kg)	<i>g</i>	Exhaust gas
<i>s</i>	Entropy, kJ/(kg.K)	<i>PR</i>	Power and refrigeration cycle
<i>ṁ</i>	Mass flow rate (kg/s)	<i>R</i>	Refrigeration cycle
<i>ē</i>	Specific energy, (kJ/Kg)	<i>0</i>	Environmental state
<i>Ė</i>	Energy rate	<i>1,2,3....12</i>	Thermodynamic state point
<i>Q̇</i>	Heat transfer rate (kW)	<i>1</i>	1st law of thermodynamics
<i>Ẇ</i>	Rate of work (kW)	<i>2</i>	2nd law of thermodynamics
<i>ī</i>	Exergy destruction (kW)	<i>P_{high}</i>	High circulating pressure
Greek Symbol		<i>tot</i>	total
<i>η</i>	Efficiency, %	<i>HRHE</i>	Heat recovery heat exchanger
<i>ε</i>	Effectiveness, %	<i>P</i>	Power cycle
<i>Δ</i>	Difference	<i>PL</i>	Low circulating pressure
Subscript		<i>ref</i>	Refrigeration cycle
<i>tur</i>	turbine	<i>thrott</i>	Throttling valve
<i>cooler</i>	Gas cooler	<i>L</i>	Low
<i>eva</i>	Evaporator	<i>net</i>	Net output
<i>fore</i>	Fore cooler	<i>out</i>	Outlet
		Superscript	
		<i>s</i>	Ideal state

conditions of the marine environment.

Currently, WHR systems for ships primarily consist of gas turbines (De Paepe et al., 2017), Rankine cycles (RC) (Liu et al., 2020), Organic Rankine cycle (ORC) (Köse et al., 2020), Kalina cycle (KC) (Eller et al., 2017), supercritical carbon dioxide Brayton cycle (SCBC) (Pan et al., 2020), and some combined cycles (Singh and Pedersen, 2016). Studies on these systems usually focus on designing different WHR system configurations based on the ship’s power requirement. Marine vessels mostly produce waste heat within the medium to low-quality temperature ranges, and its constant availability during vessel operation, its type (hot water, low-pressure steam, and exhaust gases), its varying energy intensity, and the large quantity it generates as a result of fuel consumption and vessel size are some of its key characteristics that makes the development of a more efficient marine WHR system scientifically viable (Lion et al., 2020; Zhu et al., 2020). Additionally, geographical factors like seasonal changes, sea conditions, and location affect the

quality and quantity of heat generated. Marine WHR systems need to adapt to dynamic operations for optimal performance. An effective WHR systems should seamlessly integrate with the existing ship power systems and offer short payback periods for economic viability (Zhu et al., 2020). WHR technology for marine vessels should enable increased energy savings, typically integrated with the main engine or heat-generating auxiliary components without the system being too large. Fig. 1 illustrates some the applications of WHR technologies in marine vessels.

Although the most commonly used cycle is the Rankine cycle, it has many inherent disadvantages, such as: low vapour pressure, low quality of heat transfer at the evaporator and the requirement to superheat the water before being pumped (Nguyen et al., 2010). That’s why, CO₂ is a potential substitute due to its appealing thermodynamic and physical properties, such as its lower specific volume and higher specific heat at constant pressure (Ehsan et al., 2018a). Additionally, it is non-toxic and

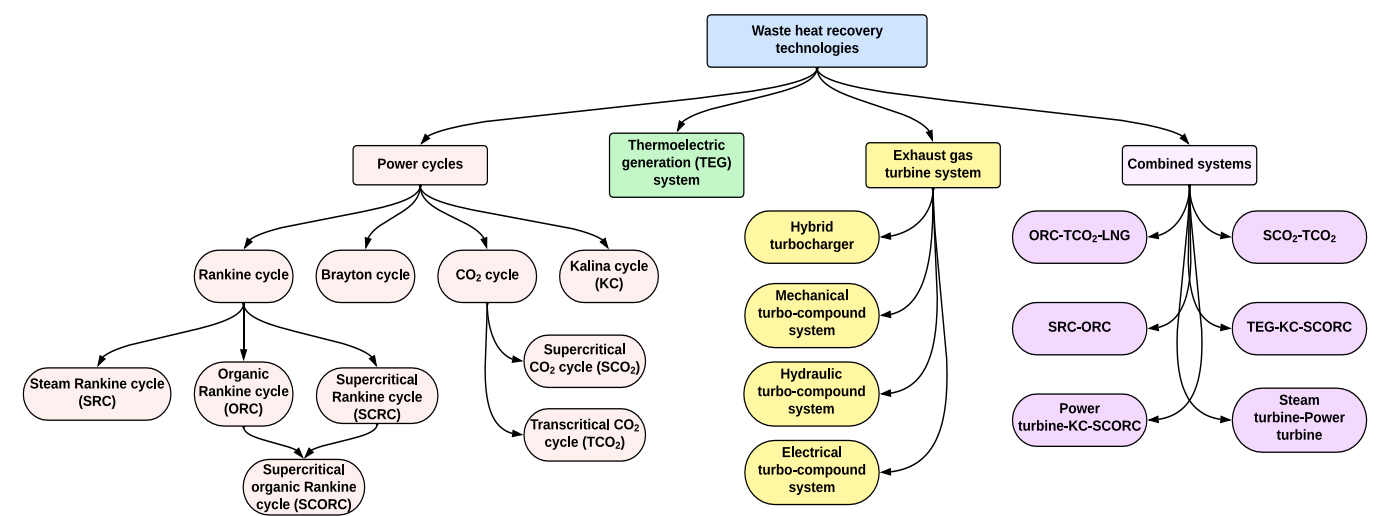


Fig. 1. Examples of WHR technologies used in marine system (Baldasso et al., 2020; Di Battista et al., 2019; Lion et al., 2020; Orr et al., 2016; Sadreddini et al., 2018).

non-flammable, and offers complete personal and environmental safety as well as efficient operation. Since CO₂'s critical temperature is near room temperature, it is also easier for it to reach its supercritical state. The necessary work is much lessened when supercritical CO₂ (S-CO₂) is close to its critical point. Additionally, the increased concentration of S-CO₂ makes it possible to construct turbomachinery that is incredibly small and suitable to use in small heat exchangers (Sarkar, 2015). To increase the efficiency of the cycle and reduce its inconsistent heat exchanger temperatures, numerous altered S-CO₂ power cycle configurations have been proposed, including the recompression cycle (Ehsan et al., 2019a; Ehsan, Duniam, Li, et al., 2019), recuperation cycle (Crespi et al., 2017) with reheating, as well as partial cooling (Ehsan et al., 2019a; Ehsan et al., 2019b; Ehsan et al., 2018b; Ehsan et al., 2020; Ehsan et al., 2018c; Ehsan et al., 2020). The enhanced performance of an appropriately modified S-CO₂ cycle underscores its potential as a power generation system. Consequently, S-CO₂ Brayton cycles have been adopted across various industries due to their significant benefits, including lower efficiency loss (M.-J. Li et al., 2019), economic feasibility (H. Li et al., 2019a; H. Li et al., 2019b; J. Li et al., 2019), and suitability for smaller power plants (Chai and Tassou, 2023).

Furthermore, for refrigeration cycles natural refrigerants are now widely used due to the negative environmental effects of conventional refrigerants. Using CO₂ in its transcritical phase as a working fluid for power generation is a challenge as very precise control of temperature is needed to avoid unwanted condensation. However, it is ideal for a refrigeration cycle. Evidently, many studies have been conducted on the transcritical CO₂ (T-CO₂) refrigeration cycle for use in the automotive, retail, and marine industries as CO₂ has beneficial refrigeration qualities such as high volumetric capacities and thermal conductivity, as well as high latent heat and for its capacity to be used as a dry cooling medium. A modified T-CO₂ cycle is reported to improve refrigeration performance substantially (B. Yu et al., 2019). In a study conducted by Bai et al. (Bai et al., 2016) on a T-CO₂ refrigeration cycle with an integrated ejector expansion system, it was discovered that the system's exergy destruction is predominately endogenous, and that by upgrading the system components, 43.44 % of the overall exergy destruction could be prevented. The dual-stage T-CO₂ cycle's effectiveness combined with two ejectors was investigated by Manjili and Cheraghi (Eskandari Manjili and Cheraghi, 2019). The outcome showed that, when compared to the traditional cycle, the altered cycle enhances coefficient of performance (COP) up to 80 % from 20 %.

With the numerous advantages in using supercritical CO₂ as a working fluid for a power cycle and transcritical CO₂ for a refrigeration cycle, it makes sense to combine them effectively to make a heat recovery system, particularly for waste heat recovery in marine vessels where there is a size limitation (Ehsan et al., 2023). Few studies have been conducted thus far for CO₂ based integrated power and refrigeration systems. Wang et al. (J. Wang et al., 2012) proposed a combined

system that integrated a Brayton cycle and a modified T-CO₂ for combined cooling, heating, and power (CCHP). Xu et al. (Xu et al., 2015) enhanced the exergy efficiency of this system by approximately 12.1 % by adding an extraction turbine. However, employing such a straightforward method for power and refrigeration results in an overall low exergy efficiency. A unique combined power and refrigeration system made up of a S-CO₂ recompression cycle and a T-CO₂ refrigeration cycle was thermodynamically studied by Li and Wang (B. Li and Wang, 2019) where the collected results showed that combined systems are more advantageous than separate systems by increasing the cooling capacity and reducing desired temperature in the evaporator, with the combined cycle showing a 5.87 % increase in exergy efficiency relative to acting as two separate cycles. Numerous studies highlight the significance of combined systems using S-CO₂ and T-CO₂ cycles for waste heat recovery, especially in marine power and cooling systems, as summarized in Table 1:

However, a severe temperature mismatch in the heat exchangers of such systems, originated by the hot and cold stream's difference in heat capacity, coupled with the general energy loss to the environment accounts for almost half of the overall irreversibility, which makes it economically unfeasible. One potential solution is to use a colder medium in the CO₂ cycle as a condenser, such as an LNG cycle (Sadreddini et al., 2018), however, this solution is not available for marine vessels as the size would increase significantly. Using a simple ORC (unmodified) as a topping cycle to harness the low-to-medium temperature is another potential solution, but it's high irreversibility and lack of a suitable bottoming cycle without substantially increasing the overall size of the system, still remains an important issue. Combined power and refrigeration systems also suffer from its inherent disadvantage of having a reduced cooling capacity compared to standalone cycles. To enhance the system's effectiveness, the addition of a flash-tank to the refrigeration cycle can be implemented (Nabil et al., 2023). This causes the liquid to "flash" into a vapor phase, which can then be separated from the other liquid. A second law thermodynamic analysis carried out by Sirwan et al. (2013a) to assess the performance of an absorption refrigeration cycle, modified with a flash-tank and an ejector, elucidated that there is a significant enhancement in the system's performance after modification with the COP increasing from 0.844 to 0.875 and the exergetic efficiency increasing from 0.459 to 0.476. Another study by the same set of authors indicated that the inclusion of a flash-tank led to the optimization of the ejector entrainment ratio and an increase in the condenser pressure with an improvement in the COP of the system (Sirwan et al., 2013b). This optimization is especially valuable in hot climates, where the efficiency of the condenser suffers due to the warm temperatures. Integrating a flash-tank also improved overall performance in cascaded refrigeration systems such as compression-absorption where their proposed system has achieved 18.49 % higher COP and exergy efficiency than conventional system (Faruque et al., 2023a) and ejector-injection (Faruque

Table 1
Overview of combined systems for waste heat recovery from literature.

Author	Power cycle	Refrigeration cycle	Working fluids	Performance				Ref.
				Net work (kW)	Energy efficiency (%)	Exergy efficiency (%)	COP	
Manjunath et al.	Supercritical CO ₂ cycle	Transcritical CO ₂ cycle	Carbon dioxide	3030	45.6	62.8	2.75	(Manjunath et al., 2018)
Cao et al.	Gas turbine combined with a supercritical CO ₂ Brayton cycle	Transcritical CO ₂ cycle	Carbon dioxide	9250	51.41	-	-	(Cao et al., 2017)
Sharma et al.	S-CO ₂ based regenerative Brayton cycle	-	Carbon dioxide	5148	39.6	62.3	-	(Sharma et al., 2017)
Choi et al.	T-CO ₂ Rankine cycle with 2-stage reheat and 3-stage turbine	-	Carbon dioxide	383	7.87	5.94	-	(Choi, 2016)
Wahab et al.	S-CO ₂ Brayton cycle integrated with ORC	Absorption refrigeration cycle	Carbon dioxide	506.72	34.92	59.21	0.54	(Mubashir et al., 2023).
Dario et al.	S-CO ₂ closed cycle	-	Carbon dioxide	5700	27.5	-	-	(Alfani et al., 2021)

et al., 2023b) where an improvement of 8.57 % in COP and 7.24 % in exergy efficiency has been noticed. These studies validate that the integration of a flash-tank can facilitate the development of a combined power and refrigeration system without worrying too much about the reduced cooling capacity, while also accounting for the varying temperature changes of a marine environment.

Therefore, a sophisticated system must be created to provide sufficient power and refrigeration without suffering from substantial energy loss. This study introduces a unique WHR system, dubbed the “Marine Energy Recovery System (MERS)” that combines a supercritical CO₂ Brayton cycle, with an ORC bottoming power cycle and a transcritical CO₂ refrigeration cycle, enhanced with a flash-tank as well as a low-pressure compressor (LPC), a mixing chamber, and throttling valves, forming a novel cascaded system for providing power and refrigeration with improved efficiency. This novel system configuration, introduces an ORC as a bottoming power cycle to reduce the irreversibility of a CO₂ combined cycle, while also increasing the effectiveness of the refrigeration cycle, over a large temperature range. Thereby, the MERS is designed to optimize waste heat recovery in marine vessels, significantly improving efficiency and practicality compared to existing systems. The integration of the flash-tank within the MERS plays a crucial role by stabilizing the pressure and enhancing the efficiency of the refrigeration cycle, resulting in better overall system performance. The inclusion of ORC allows for the utilization of more waste heat from the system, converting it into useful work, further enhancing the energy efficiency and exergy efficiency of the entire setup. The ORC effectively captures lower-grade waste heat that would otherwise be lost, maximizing the energy recovered from the marine shipboard engine. Moreover, a Multi-Objective Optimization (MOO) is also conducted using Genetic Algorithm. The objective function is generated by an Artificial Neural Network (ANN) model from the relationship between the input variables and multiple objectives. The MOO algorithm will then utilize the trained model to explore the optimal conditions. The gas cooler pressure, the gas cooler outlet temperature and turbine inlet temperature have been taken as the input parameters, while the objective values are the net work and

the energy efficiency of the system. With the added advantage of CO₂-based cycles being smaller in size, the MERS can be utilized for all types of marine vessels. By addressing these points, this study distinguishes itself as a novel cascaded system through its unique system configuration, enhanced efficiency strategies, comprehensive analysis, and practical application focus, thereby making a significant contribution to the field of waste heat recovery in maritime applications.

2. System description

The system proposed by Yu et al. (2020) outlined the usage of recuperators in a CO₂ power and refrigeration system, as can be seen in Fig. 2. Yet, some heat is still lost in the environment, which can be recovered and used in another power cycle. With that thought, a new configuration has been proposed in this paper. The reference system is a system proposed by Manjunath et al. (2018) to harness the waste heat generated from shipboards through the utilization of recuperators, which was further modified by Yu et al. (2020).

2.1. Conventional power and refrigeration system

Fig. 2 illustrates the WHR system that was proposed by Yu et al. (2020). It operates by generating power and refrigeration simultaneously by utilizing the waste heat. The power circuit consists of the heat-recovering heat exchanger (HRHE), a turbine, a high-temperature recuperator (HTR), a low-temperature recuperator (LTR), a gas cooler, and a high-pressure compressor (HPC). The power circuit is run in the pressure range where CO₂ is constantly in its supercritical phase. The components of the refrigeration circuit are the LTR, gas cooler, forecooler, throttling valve, evaporator and low-pressure compressor (LPC). In the refrigeration circuit, to ensure efficient compression of CO₂, the supercritical state is achieved after the stream passes through the LPC and, during throttling and the evaporation process, CO₂ remains in its subcritical phase. It is named as the “Transcritical CO₂ cycle” for this reason.

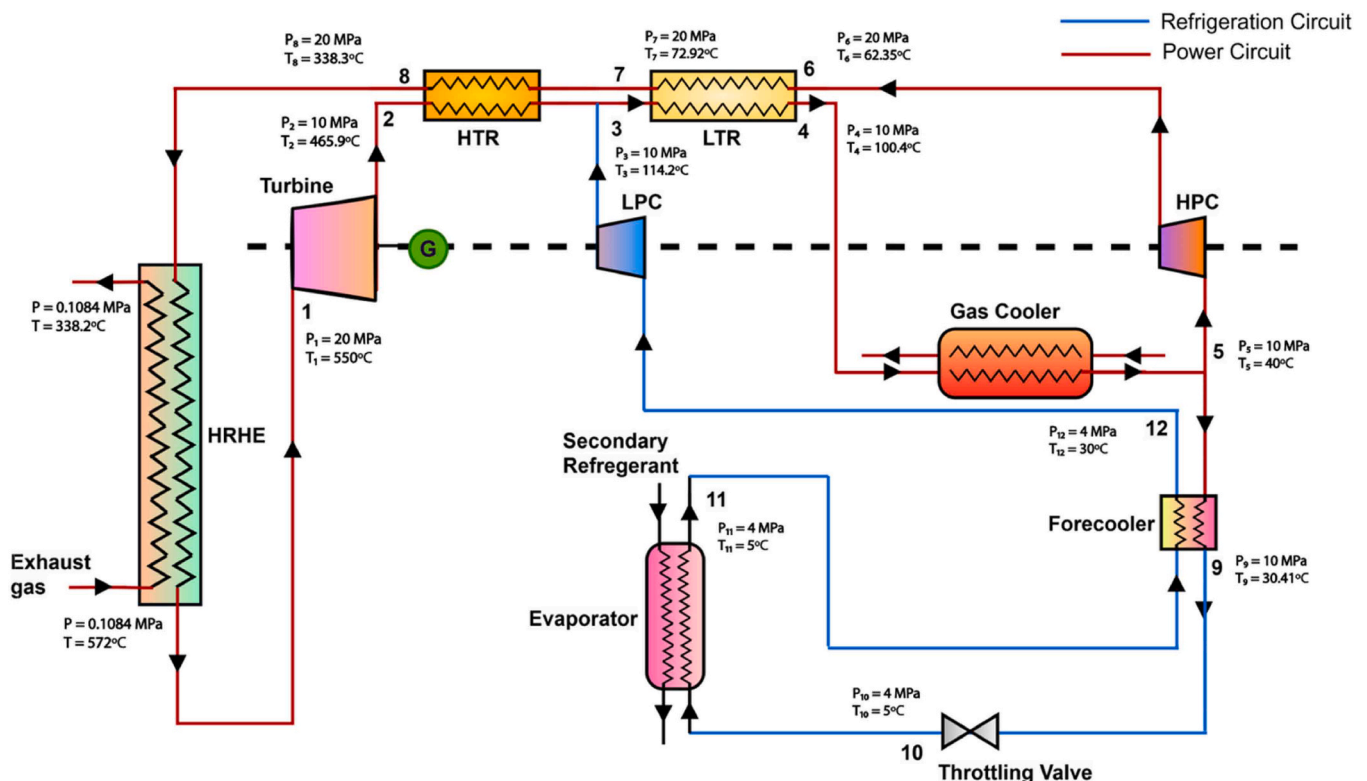


Fig. 2. Conventional cooling and power system proposed by Yu et al. (2020).

In this system, at state point 1, the S-CO₂ absorbs the exhaust gas heat from the gas turbine through the HRHE, which causes the CO₂ temperature to rise. CO₂, at high temperature, enters the turbine (state point 1), and goes through isentropic expansion to state point 2 to produce work. The hot CO₂ from the turbine then gives out heat to the high-pressure cold flow as they travel through the HTR. At the HTR's output (state point 3), the CO₂ flow from the power circuit and the flow from the LPC of the refrigeration circuit merge. In order to transfer heat to the HPC exit flow, the heated mixed CO₂ next enters the LTR at state point 4. The CO₂ then enters through the gas cooler to release heat and cools at state point 5. After that, the CO₂ stream is used to create a power stream and a refrigeration stream.

In the power stream, HPC compresses the flow to state point 6 where it then takes up heat from the LTR and HTR (state point 7 & 8, respectively) before going back into the HRHE. Ultimately, the power circuit is finished when the heated S-CO₂ enters the HRHE. On the other hand, the refrigeration stream travels to the forecooler (state point 9) where it rejects heat by superheating the outlet CO₂ from the evaporator. The employment of a forecooler can enhance the thermodynamic efficiency of the system, especially for systems that employ S-CO₂ as their working fluid (Ahmed and Ehsan, 2023). The stream enters the throttling valve at the end of state point 9, where it exits the forecooler. Further cooling and a lower pressure are achieved during the throttling process from state 9 to state 10. The stream then moves into the evaporator, where it produces cooling by absorbing heat from its surroundings and reaches state point 11. The forecooler superheats the evaporator's outlet flow before it exits into the LPC. The refrigeration stream then combines with the power stream again at state point 3.

2.2. Proposed combined integrated power and refrigeration system

The schematic of the proposed “Marine Energy Recovery System

(MERS)”, is shown in Fig. 3, highlighting the design of cycle configurations and the state points. This configuration comprises of a supercritical CO₂ (S-CO₂) power circuit and a transcritical CO₂ (T-CO₂) refrigeration circuit for simultaneous power and refrigeration.

In addition to the six fundamental components (i.e., gas turbine, HTR, LTR, gas cooler, HPC and HRHE) in the power circuit, an Organic Rankine Cycle (ORC) has been introduced to recover the dissipated heat, which uses Butane as its working fluid, which is reported to improve performance when a low-grade heat source, such as the heat rejected in the gas cooler, is used (Song et al., 2020). The HTR and LTR are used for recuperation of heat. Even though studies elucidate that, for other cycles such as: optimizing the ammonia mass fraction of the Kalina cycle (KC) and considering the fluid phase after the evaporator adds flexibility and enhanced efficiency when used as a WHR system (Eller et al., 2017), practical implementation must address safety, chemical resistance, and oxidation concerns, which makes its integration a tough and expensive choice (Zhu et al., 2020). However, Organic Rankine Cycle has been chosen for this system. In the ORC, the gas cooler works as a boiler by exchanging sufficient heat with the main power circuit, and the stream flows through the gas cooler through the turbine at state points 18–19, where it expands to a lower pressure, producing work. It then enters the condenser, releasing heat at state points 19–20, before being pumped to a higher pressure to state point 21 and entering into the gas cooler again. Just like the conventional cycle, the CO₂ power stream in the MERS is in its supercritical state.

The T-CO₂ refrigeration circuit consists of the previous LTR, gas cooler, evaporator, forecooler. However, two low-pressure compressors (LPC-I, LPC-II) are used, along with two throttling valves, a flash-tank and a mixing chamber. In this enhanced refrigeration cycle, similar to the reference system, the supercritical state is achieved after the refrigeration stream passes through LPC-I while sub-critical state is maintained in the rest of the refrigeration circuit components. Just like

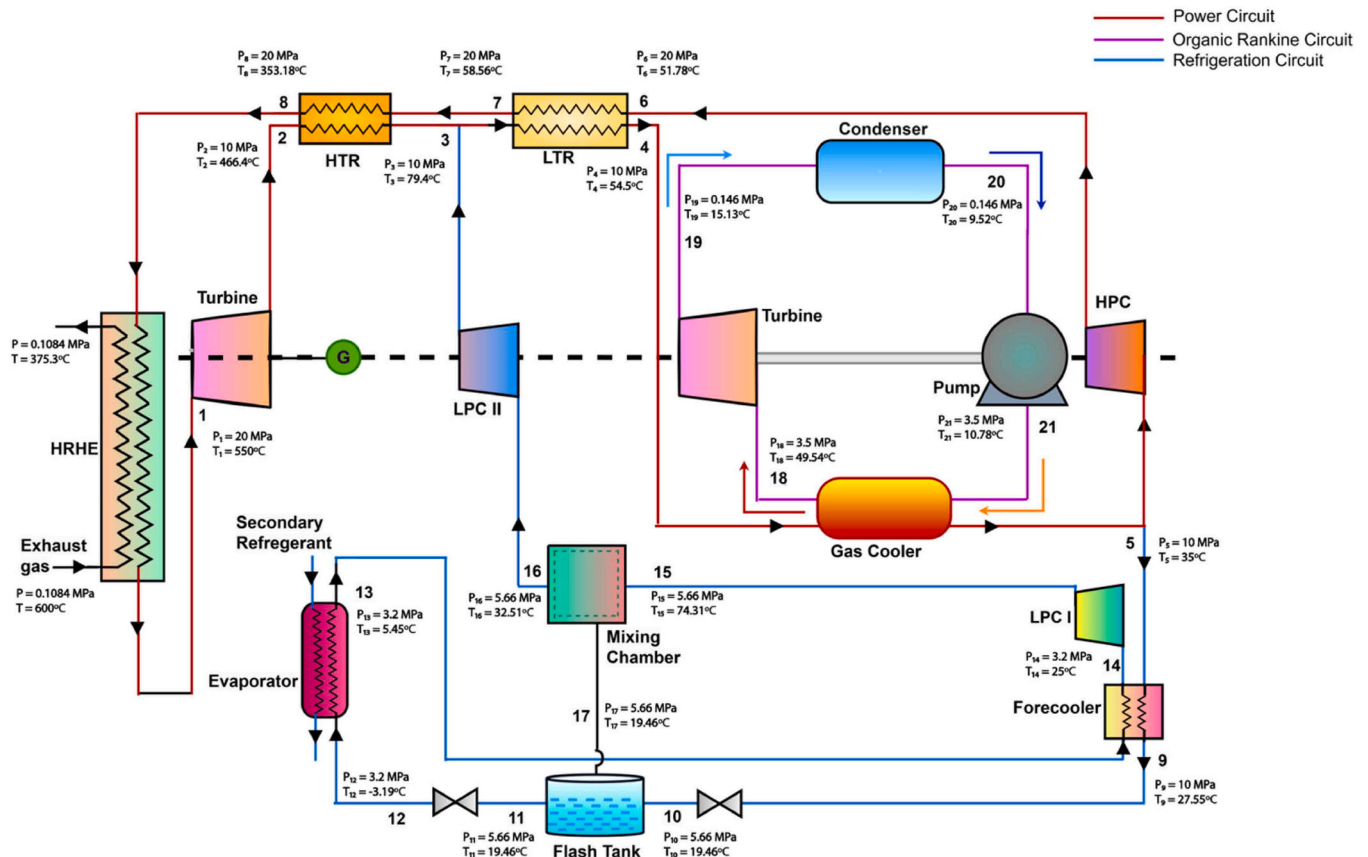


Fig. 3. Proposed integrated Power and Refrigeration System for MERS application.

the reference system, the refrigeration flow is routed to the forecooler with the aim of superheating CO₂. Afterward, it exits the forecooler at state point 9 and proceeds to the initial throttling valve. During the initial throttling process (state 9–10), CO₂ is dropped to an intermediate pressure, while simultaneously cooling down. The stream then travels from state point 10 to state point 11, through the flash-tank, where the gaseous portion of the stream is separated and flows to state point 17 to the mixing chamber. The liquid portion of the CO₂ stream goes through the second throttling value, to achieve a lower pressure and temperature at state point 12. Then, the flow goes through the evaporator where the main refrigeration effect happens by the absorption of heat from the surrounding, to point 13. Before entering the LPC-I, the outflow from the evaporator is subjected to superheating within the forecooler. Finally, the CO₂ is mixed with the “flashed” gaseous part of the flow in the mixing chamber at state point 15 and leaves as a combined stream to state 16 to the LPC - II and then again mixes with power circuit stream at point 3, thus accomplishing the refrigeration circuit. The temperature versus entropy diagram is depicted in Fig. 4.

3. Thermodynamic modelling

State characteristics and performance parameters of each component in the corresponding cycle can be calculated using mass, energy, and exergy balance equations.

3.1. Energy analysis

Each of the parts can be thought of as a control volume system, and the mass balance equation can be created as follows by using the continuity equations which is given in Eq. (1) (Çengel et al. 2019).

$$\sum \dot{m}_i = \sum \dot{m}_e \quad (1)$$

where, i and e refers to the inlet and outlet flow of the system, respectively.

The steady flow energy equation (SFEE), which is derived by applying the first rule of thermodynamics, is used to calculate the total energy flow through the systems as follows in Eq. (2): (Çengel et al. 2019)

$$\sum \dot{m}_i \left(h_i + \frac{V_i^2}{2} + gz_i \right) + \dot{Q} + \dot{W} = \sum \dot{m}_e \left(h_e + \frac{V_e^2}{2} + gz_e \right) + \dot{W} \quad (2)$$

The formulation omits potential and kinetic energy; the equation (Eq. (3)) is condensed to: (Ghaebi et al., 2017a)

$$\sum \dot{m}_i h_i + \dot{Q} = \sum \dot{m}_e h_e + \dot{W} \quad (3)$$

Where $\dot{m}$ is the mass flow rate, $\dot{Q}$ is the heat transfer rate, and $\dot{W}$ is the power. The turbine efficiency follows the following Eq. (4) (Çengel et al. 2019),

$$\eta_{tur} = (h_{in} - h_{out}) / (h_{in} - h_{out,s}) \quad (4)$$

where, ‘s’ refers to the ideal state of the turbine and ‘in’ and ‘out’ refers to the inlet and outlet of the flow of the system, respectively.

The net power output of the system can be determined by this equation (Eq. (5)) (A. Yu et al., 2020):

$$\dot{W}_{net} = \dot{W}_{tur} - \dot{W}_{HPC} - \dot{W}_{LPC} \quad (5)$$

where, $\dot{W}_{tur}$, $\dot{W}_{HPC}$ and $\dot{W}_{LPC}$ is the work output from the turbine, work input from the High-Pressure Compressor (HPC) and the work input required by the low-pressure compressor (LPC) in the refrigeration cycle, respectively.

Eq. (6) gives the energy efficiency of the power cycle (Ghaebi et al., 2017b):

$$\eta_{1,p} = \frac{(\dot{W}_{tur} + \dot{W}_{ORC} - \dot{W}_{HPC})}{\dot{Q}_{in}} \quad (6)$$

For the ORC the generated power by turbine can be calculated using the following Eq. (7) (Sadreddini et al., 2018)

$$\dot{W}_{tur} = \dot{m}_{in} (h_{in} - h_{out}) \quad (7)$$

The thermodynamic efficiency can be found from Eq. (8),

$$\eta_{ORC} = \dot{W}_{ORC} / \dot{Q}_{ORC,in} \quad (8)$$

The coefficient of performance which evaluates the corresponding performance of the refrigeration system can be determined by Eq. (9) (Ghaebi et al., 2017b),

$$COP = \frac{\dot{Q}_{ref}}{\dot{W}_{LPC}} \quad (9)$$

Finally, energy efficiency of the MERS can be found from the following Eq. (10),

$$\eta_{PR} = \frac{(\dot{W}_{net} + \dot{W}_{ORC} + \dot{Q}_{ref})}{\dot{Q}_{in}} \quad (10)$$

The energy efficiency of the HRHE, i.e., efficiency of waste heat recovery can be found using Eq. (11)

$$\eta_{1,WHR} = \frac{\dot{Q}_{in}}{[m_{g,in} (h_{g,in} - h_0)]} \quad (11)$$

where, ‘0’ refers to the environmental state of the system.

3.2. Exergy Analysis

Exergy is the maximum amount of work that a system is capable of producing when it achieves equilibrium with its environment. The exergy can be defined as follows with Eq. (12) in accordance with the second law of thermodynamics: (Çengel et al., 2011)

$$\dot{E}x = \dot{m}_r [(h - h_0) - T_0 (s - s_0) + \frac{V^2}{2} + gz] \quad (12)$$

The formula can be made simpler by ignoring the impact of kinetic and potential energy on the exergy as (Eq. (13)): (Adrian Bejan et al., 1995)

$$\dot{E}x = \dot{m}_r [(h - h_0) - T_0 (s - s_0)] \quad (13)$$

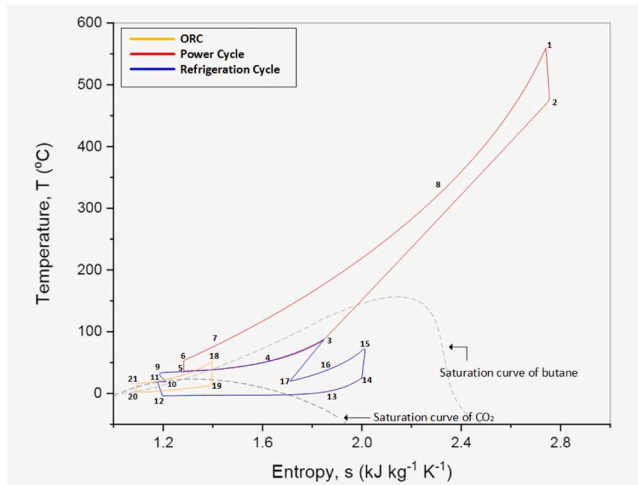


Fig. 4. Temperature-entropy (T-s) diagram of the proposed system (MERS).

Or, for this system it can be written,

$$e_i = h_i - h_0 - T_0(s_i - s_0) \quad (14)$$

The subscript 'i' refers to the state points of each component that makes up the system. On the other hand, the subscript '0' pertains to the environmental state. The overall exergy input rate for the power and refrigeration systems is determined by Eq. (15) (A. Yu et al., 2020),

$$E_{in,PR} = Q_{in}(1 - T_0/T_{g,ave}) \quad (15)$$

Here, $T_{g,ave}$ describes the average temperature of the waste exhaust gas. Eq. (16) illustrates how to compute $T_{g,ave}$ by finding the ratio of the enthalpy and entropy difference between the exhaust gas inlet and exhaust gas outlet (Ghaebi et al., 2017b).

$$T_{g,ave} = \frac{(h_{g,in} - h_{g,out})}{(s_{g,in} - s_{g,out})} \quad (16)$$

The refrigeration cycle's useful exergy output is found using Eq. (17): (A. Yu et al., 2020)

$$E_R = |Q_{ref} \left(1 - \frac{T_0}{T_{r,ave}}\right)| \quad (17)$$

where the refrigerated space's temperature average is calculated by Eq. (18) (Ghaebi et al., 2017b):

$$T_{r,ave} = \frac{(h_{r,in} - h_{r,out})}{(s_{r,in} - s_{r,out})} \quad (18)$$

The exergy output rate of the power system is found by Eq. (19):

$$E_p = \dot{W}_{net} + \dot{W}_{ORC} \quad (19)$$

The power and refrigeration cycle's exergy rate adds up to the system's net exergy output. Consequently, the system's exergy efficiency can be determined by Eq. (20),

$$\eta_{2,PR} = \frac{(E_R + W_{net})}{E_{in,PR}} \quad (20)$$

The recovery efficiency of waste heat is determined using the second law of thermodynamics as follows in Eq. (21) (A. Yu et al., 2020):

$$\eta_{2,WHR} = \frac{(e_{g,in} - e_{g,out})}{[h_{g,in} - h_0 - T_0(s_{g,in} - s_0)]} \quad (21)$$

4. Mathematical modelling

The mathematical model is developed to evaluate the performance of the MERS. Assumptions are applied across each component taken as a control volume to develop the thermodynamic model of the proposed combined cycle to simulate the interrelation between the state properties as follows (Bishal et al., 2023; Khan et al., 2023; Faruque et al., 2022; Yu et al., 2020).

- All components are expected to operate under steady-state conditions.
- Any internal heat transfer or pressure drops through the piping is deemed negligible, hence ignored.
- All the heat exchangers calculations are considered counter-flow.
- Constant isentropic efficiencies serve as a representation for expansion and compression processes.
- It is presumed that fluid properties will be uniform across the cross-section at the mixing chamber's exit.
- The working fluid of the heat source is exhaust air, and is regarded as an ideal gas.

4.1. Modelling of the CO₂ Power and Refrigeration circuits of the MERS

The mathematical model developed for the system and their relevant equations for each component are given below:

Heat Recovery Heat Exchanger (HRHE): Eq. (22) gives,

$$T_{g,out} = T_8 + \Delta T_{HRHE} \quad (22)$$

where, $T_{g,out}$ is the output temperature of the exhaust gas of the gas turbine.

Turbine: The state point 1 and 2 refer to the turbine inlet and outlet respectively. Here Eq. (23) states,

$$s_{2s} = s_1 \quad (23)$$

The turbine efficiency is represented by the following Eq. (24),

$$\eta_{turb} = \frac{(T_1 - T_2)}{(T_1 - T_{2s})} \quad (24)$$

High-Temperature Recuperator (HTR): It is a component that is designed to recover and utilize high-temperature waste heat in a system. It transfers thermal energy from hot fluids (exhaust gas) to a working fluid (CO₂) as a heat exchanger. Thus, Eq. (25) becomes,

$$h_8 - h_7 = h_2 - h_3 \quad (25)$$

The effectiveness of the heat exchanger follows the following Eq. (26),

$$\varepsilon_{HX} = \frac{(h_3 - h_7)}{(h_3 - h_6)} \quad (26)$$

Low-Temperature Recuperator (LTR): It is also designed to recover and utilize low-temperature waste heat and it transfers thermal energy from relatively cooler gases or fluids to the working fluid of the system as a heat exchanger. LTR output state can be calculated by Eq. (27),

$$h_{7,PL} = m_{CO2,P}(h_7 - h_6) \quad (27)$$

High-Pressure Compressor (HPC): HPC increases the pressure of the working fluid, by bringing it in at a lower pressure and compressing it to a higher pressure. The properties for the HPC can be found from Eq. (28):

$$\eta_{HPC} = \frac{(h_{6s} - h_5)}{(h_6 - h_5)} \quad (28)$$

Low-Pressure Compressor (LPC): The function of LPC-I is to increase the pressure of the refrigeration circuit's working fluid to match the pressure of the gaseous fluid that was removed from the flash-tank. LPC-II increases the pressure of the refrigeration circuit's working fluid further to match the supercritical pressure of the power circuit's working fluid. The properties for the corresponding state points can be calculated from Eqs. (29–30).

$$\eta_{LPC,I} = \frac{(t_{15s} - t_{14})}{(t_{15} - t_{14})} \quad (29)$$

$$\eta_{LPC,II} = \frac{(h_{3s} - h_{16})}{(h_3 - h_{16})} \quad (30)$$

Gas cooler: For the gas cooler, the inlet (state 4), the pressure is the same as the LTR hot side inlet pressure (state 3) as, in the recuperator, the pressure remains unchanged which can be seen in Eq. (31),

$$P_4 = P_3 \quad (31)$$

From the gas cooler, the flow gets divided into two ways, one way leads to the power cycle where state 6 and 7 represents the inlet and outlet of the LTR cold side. Here Eqs. (32–35) clearly represents the conditions,

$$if T_7 \leq T_6 \quad (32)$$

$$\text{At the LTR : } T_7 = T_6 \quad (33)$$

$$T_4 = T_3 \text{ and } h_6 = h_7 \quad (34)$$

$$\text{else } T_4 = T_3 - \varepsilon_{HX}(T_3 - T_6) \quad (35)$$

Fore cooler: The other part of the flow after division at the gas cooler outlet, flows through the fore cooler and enters into the refrigeration cycle which follows the conditions below in Eq. (36),

$$h_5 - h_9 = h_{14} - h_{13} \quad (36)$$

Throttling valves: The temperature of the working fluid remains constant during the throttling process and can be expressed by Eq. (37) (A. Yu et al., 2020).

$$h_{10} = h_9 \text{ and } T_{11} = T_{10} \quad (37)$$

Evaporator: The change in temperature in the evaporator has been kept constant in this system as the previous study (A. Yu et al., 2020). Thus, the conditions for this state are presented by Eq. (38):

$$P_{12} = P_{13}, \quad T_{12} = T_{13} \text{ and } h_{12} = h_{11} \quad (38)$$

The equation that is used to calculate the mass flow rate of the power cycle is Eq. (39):

$$\dot{m}_{CO2,P} = \frac{\dot{Q}_{in}}{(h_1 - h_8)} \quad (39)$$

The energy, mass balance and rate of exergy destruction equations for main components in the power and refrigeration section of the system, according to the 1st and 2nd law of thermodynamics, are respectively presented in below Table 2.

Table 2

Mass, energy balance and exergy destruction equation across each component of the power and refrigeration cycle of MERS (Çengel et al. 2019; Dincer and Zamfirescu 2014).

Components	Mass Balance	Energy Balance	Exergy Destruction
HRHE	$\dot{m}_1 = \dot{m}_8$	$\dot{Q}_{in} = \dot{m}_g(h_{g,in} - h_{g,out})$	$\dot{I}_{HRHE} = \dot{m}_{CO2,P}(\dot{e}_8 - \dot{e}_1) + \dot{m}_g(\dot{e}_{g,in} - \dot{e}_{g,out})$
Turbine	$\dot{m}_2 = \dot{m}_1$	$\dot{W}_{tur} = \dot{m}_{CO2,P}(h_1 - h_2)$	$\dot{I}_{tur} = \dot{m}_{CO2,P}(\dot{e}_1 - \dot{e}_2) - \dot{W}_{tur}$
HTR	$\dot{m}_3 = \dot{m}_2 + \dot{m}_{16}$	$\dot{Q}_{HTR} = \dot{m}_{CO2,P}(h_8 - h_7)$	$\dot{I}_{HTR} = \dot{m}_{CO2,P}[(\dot{e}_2 - \dot{e}_3) + (\dot{e}_7 - \dot{e}_8)]$
LTR	$\dot{m}_2 + \dot{m}_{16} + \dot{m}_3$	$\dot{Q}_{LTR} = \dot{m}_{CO2,P}(h_7 - h_6)$	$\dot{I}_{LTR} = \dot{m}_{CO2,P}(\dot{e}_6 - \dot{e}_7) + (\dot{m}_{CO2,P} + \dot{m}_{CO2,R})(\dot{e}_3 - \dot{e}_4)$
Gas Cooler	$\dot{m}_5 = \dot{m}_4; \dot{m}_5 = \dot{m}_9 + \dot{m}_6$	$\dot{Q}_{out} = (\dot{m}_{CO2,P} + \dot{m}_{CO2,R})(h_4 - h_5)$	$\dot{I}_{cooler} = (\dot{m}_{CO2,P} + \dot{m}_{CO2,R})(\dot{e}_4 - \dot{e}_5) - \dot{Q}_{out}(1 - T_0/T_5)$
HPC	$\dot{m}_6 = \dot{m}_5$	$\dot{W}_{HPC} = \dot{m}_{CO2,P}(h_6 - h_5)$	$\dot{I}_{HPC} = \dot{m}_{CO2,P}(\dot{e}_5 - \dot{e}_6) + \dot{W}_{HPC}$
Fore cooler	$\dot{m}_9 = \dot{m}_5$	$\dot{Q}_{fore} = \dot{m}_{CO2,R}(h_5 - h_9)$	$\dot{I}_{fore} = \dot{m}_{CO2,R}[(\dot{e}_5 - \dot{e}_9) + \dot{m}_{liq}(\dot{e}_{13} - \dot{e}_{14})]$
Evaporator	$\dot{m}_{13} = \dot{m}_{12}$	$\dot{Q}_{ref} = \dot{m}_{liq}(h_{13} - h_{12})$	$\dot{I}_{eva} = \dot{m}_{CO2,R}(\dot{e}_{12} - \dot{e}_{13}) + \dot{E}_{R,PR}$
Throttling valve	$\dot{m}_{12} = \dot{m}_{11}; \dot{m}_{10} = \dot{m}_9$		$\dot{I}_{thrott} = \dot{m}_{CO2,R}(\dot{e}_9 - \dot{e}_{10}) + (\dot{e}_{11} - \dot{e}_{12})$
Flash-tank	$\dot{m}_{11} = \dot{m}_{10}$		$\dot{I}_{flash} = \dot{m}_{CO2,R}\dot{e}_{10} - \dot{m}_{liq}\dot{e}_{11} - \dot{m}_{vap}\dot{e}_{17}$
Mixing Chamber	$\dot{m}_{16} = \dot{m}_{15} + \dot{m}_{17}$		$\dot{I}_{mixc} = \dot{m}_{vap}\dot{e}_{17} + \dot{m}_{liq}\dot{e}_{15} - \dot{m}_{CO2,R}\dot{e}_{16}$
LPC - I	$\dot{m}_{15} = \dot{m}_{13}$	$\dot{W}_{LPC,I} = \dot{m}_{liq}(h_{15} - h_{14})$	$\dot{I}_{LPC,I} = \dot{m}_{CO2,R}(\dot{e}_{14} - \dot{e}_{15}) + \dot{W}_{LPC,I}$
LPC - II	$\dot{m}_3 = \dot{m}_{16} + \dot{m}_2$	$\dot{W}_{LPC,II} = \dot{m}_{CO2,R}(h_3 - h_{16})$	$\dot{I}_{LPC,II} = \dot{m}_{CO2,R}(\dot{e}_{16} - \dot{e}_3) + \dot{W}_{LPC,II}$
LPC		$\dot{W}_{LPC} = \dot{W}_{LPC,I} + \dot{W}_{LPC,II}$	$\dot{I}_{LPC} = \dot{I}_{LPC,I} + \dot{I}_{LPC,II}$

4.2. Modelling of the ORC section of the MERS

For the ORC, the following equations are used to find the thermodynamic properties of each component. Here the gas cooler of our cycle works as the boiler, and the generated heat can be calculated by the Eqs. (40, 42) which are given below.

$$\dot{Q}_{Cooler} = (\dot{m}_{CO2,P} + \dot{m}_{CO2,R})(h_4 - h_5) \quad (40)$$

$$\dot{Q}_{ORC,in} = \dot{Q}_{cooler} \quad (41)$$

The mass flowrate of the system is found from Eq. (42),

$$\dot{m}_{ORC} = \dot{Q}_{Cooler}/(h_{18} - h_{21}) \quad (42)$$

For the turbine at state 18, as mentioned before, the gas cooler works as a heat exchanger for the power system and as it works as the boiler for the ORC system.

$$T_{18} = T_4 - \Delta T_{GC} \quad (43)$$

where, ΔT_{GC} is the minimum temperature difference of the gas cooler.

At state 19, for the turbine outlet, isentropic expansion happens. Thus, the state point properties can be found from Eq. (44 - 47),

$$s_{19} = s_{18} \quad (44)$$

$$h_{19} = h_{18} - \eta_{tur,ORC}(h_{18} - h_{19s}) \quad (45)$$

For the pump at state 20,

$$P_{20} = P_{19} \quad (46)$$

For state 21,

$$P_{21} = P_{18} \text{ and } s_{21} = s_{20} \quad (47)$$

The mass flow rate, energy balance and exergy destruction equations for each of the components of the ORC of the MERS is given in Table 3.

4.3. System flowchart and input parameters

The reference system's cycle parameters are acquired and contrasted with the outcomes of Yua et al. to validate the proposed integrated model (A. Yu et al., 2020). The heat source specifications and input parameters that specify the baselines for the model's validation and performance assessment are given in Table 4. The flowchart in Fig. 5 illustrates the sequential steps employed in the thermodynamic model to calculate the system parameters. Python was employed to develop the corresponding program, and the property calculation of CO₂ was conducted using COOLPROP module, which was installed into the Python codes. The minimum pinch temperature for the HTR and LTR was assumed to be greater than 5°C. The isentropic efficiencies for the turbine, HPC, LPC and the effectiveness of the heat exchanger for the system are listed as 90 %. While achieving 90 % efficiency for the components might seem optimistic, there are several reasons why these values are considered reasonable within the scope of this study. Advanced turbines, especially those used in high-performance applications, can achieve isentropic efficiencies close to 90 % (A. Yu et al., 2020). Compressors, particularly in power generation applications, can achieve isentropic efficiencies nearing 90 % (Arqam et al., 2024) and

Table 3

Mass and energy balance equation across each component of ORC of MERS.

Components	Mass Balance	Energy Balance Model
Gas cooler	$\dot{m}_{18} = \dot{m}_{ORC}$	$\dot{Q}_{Cooler} = (\dot{m}_{CO2,P} + \dot{m}_{CO2,R})(h_4 - h_5)$
ORC Turbine	$\dot{m}_{19} = \dot{m}_{18}$	$\dot{W}_{tur,ORC} = \dot{m}_{ORC}(h_{18} - h_{19})$
ORC Condenser	$\dot{m}_{20} = \dot{m}_{19}$	$\dot{W}_{tur,ORC} = \dot{m}_{ORC}(h_{19} - h_{20})$
ORC Pump	$\dot{m}_{21} = \dot{m}_{18}$	$\dot{W}_{pump,ORC} = \dot{m}_{ORC}(h_{21} - h_{20})$
ORC net-work output	$\dot{m}_3 = \dot{m}_2 + \dot{m}_{16}$	$\dot{W}_{ORC,net} = \dot{W}_{tur,ORC} - \dot{W}_{pump,ORC}$

Table 4

Assumed operating parameters for the MERS (Manjunath et al., 2018; X. Wang and Dai, 2016).

Parameters	Values
Mass flow rate of heat source	63.6 kg/s
Outlet temperature of heat source	600 °C
Outlet pressure of heat source	0.1084 MPa
Temperature of the ambient	35 °C
Pressure of the ambient	0.101325 MPa
Working fluid of heat source	Air
Composition of heat source	Ideal gas
Turbine inlet pressure	20 MPa
HPC inlet pressure	10 MPa
LPC inlet pressure	4 MPa
Inlet temperature of turbine	550 °C
Inlet temperature of HPC	35 °C
Evaporator outlet temperature	5.45 °C
Evaporator outlet pressure	3.2 MPa
Forecooler pinch	10 °C
HTR minimum pinch	≥ 5 °C
LTR minimum pinch	≥ 5 °C
Evaporation and refrigerated space temperature difference	13 °C
Isentropic efficiency of the turbine	90 %
Isentropic efficiency of the HPC	90 %
Isentropic efficiency of the LPC	90 %
Heat exchanger effectiveness	90 %
Cooling cycle's CO ₂ mass flow rate	20 kg/s
CO ₂ critical temperature	31.1 °C
CO ₂ critical pressure	7.38
Operating conditions for the ORC	
Organic Rankine Cycle fluid	Butane
Gas cooler pinch, ΔT_{GC}	5 °C
Gas cooler outlet pressure	3.5 MPa
Turbine outlet pressure	146 kPa
ORC turbine efficiency, $\eta_{tur,ORC}$	86 %

modern heat exchangers designed for high performance can achieve effectiveness up to 90 % (Fakheri, 2003).

5. Model validation

To affirm the reliability of our research, the proposed integrated system, the MERS has been validated and compared with the study conducted by Yu et al. (2020)) which examined the utilization of S-CO₂ cycles and T-CO₂ refrigeration cycles for recovering waste heat in shipboards, shown in Table 5. During the validation process, specific parameters were employed: $\dot{m}_{CO_2,R} = 20$ kg/s, $P_{low} = 10$ MPa, $P_{high} = 20$ MPa, $T_{HPC\ inlet} = 40^\circ\text{C}$, $T_{evp} = 5.45^\circ\text{C}$, $\eta_{tur} = 90\%$, $\eta_{HPC} = 85\%$, $\eta_{LPC} = 80\%$, $\varepsilon_{HX} = 86\%$. To solidify the model, exergy analysis has also been conducted which is given below in Table 6.

Table 5 and Table 6 clearly demonstrate that our model exhibits a commendable level of consistency with the findings of Yua et al. (A. Yu et al., 2020). In Fig. 6, the minimal disparity between the validated and reference systems' performance is illustrated, indicating its suitability for subsequent calculations. Nevertheless, it is worth mentioning that some slight disparities between our model and the reference work can be seen. The disparities can be attributed to the reference study using MATLAB to find the state point parameters of their thermodynamic system, and using the REFPROP whereas our model is constructed in Python, utilizing its CoolProp library.

6. Results and discussion

After validation, a comprehensive thermal analysis has been conducted to assess the overall improvements in performance achieved by the MERS. This comparative analysis is based on various performance parameters, such as Coefficient of Performance (COP), net work ($\dot{W}_{net}$), net cooling ($\dot{Q}_{evp}$), energy efficiency ($\eta_{PR,1}$), exergy efficiency ($\eta_{PR,2}$), and total exergy destruction (I_{total}). Four controllable variables are considered as working parameters for our analysis: gas cooler pressure (P_{low}) ranging from 9000 kPa to 10000 kPa, evaporator pressure (P_{evp}) ranging

from 1.5 MPa to 3.5 MPa, HPC inlet temperature ($T_{HPC,in}$) set at 35°C, 40°C, 45°C, and 50°C, and evaporator temperature (T_{evp}) set at 5°C, 10°C, 15°C, and 20°C. Additionally, the effectiveness of the heat exchanger (ranging from 60 % to 95 %) and the efficiencies of the turbine (60–95 %) and compressors (65–95 %) were also considered to observe the performance of the system. Subsequently, a multi-objective optimization using genetic algorithm (GA) is performed for the MERS to identify suitable operating combinations under various working conditions. This optimization process aims to identify the most efficient and effective operating conditions for the system.

6.1. Thermodynamic performance analysis

The performance analysis was conducted using Python. To define the system, the operating parameters listed in Table 4 were assumed. For clarity, Table 7 presents the thermodynamic state parameters of the novel integrated power and refrigeration system.

6.2. Comparison between the MERS and reference system

A performance comparative analysis is conducted between the MERS and the reference system, incorporating both the first law and second law analyses at specific working conditions as outlined in Table 4. For the MERS, we have considered highly efficient components in order to get better results. The outcomes of this analysis are presented in Table 8 and Table 9. Notably, our findings reveal that the MERS enhances system energy performance by approximately 28.22 %, whereas the exergy efficiency also increases 39.06 % which is an impressive performance improvement compared to the conventional reference system.

6.3. Impacts of gas cooler pressure

In the MERS, the gas cooler plays a critical role in maintaining a pressure of 10 MPa at state 4–5. This pressure level serves as both the lowest pressure point in the power cycle and the highest pressure point in the refrigeration cycle. Consequently, this parameter holds immense importance for the entire system. Modulating this pressure value has a substantial impact on various other parameters, including net work performance, net cooling capacity, COP, energy efficiency, exergy efficiency, as well as the first and second thermodynamic waste heat recovery efficiencies of the system. Fig. 6 to Fig. 9 illustrate the diverse trends exhibited by the system's parameters as the gas cooler pressure varies within the range of 9000 kPa to 10,000 kPa. While broader pressure ranges were initially considered, our comprehensive thermodynamic analysis guided us to focus on the specified range to ensure system stability, practical feasibility, and optimal performance. The flash-tank's performance limitations at pressures beyond 10 MPa further justified our selected operating range. Thus, the trends have been investigated in relation to four different gas cooler outlet temperatures: 35°C, 40°C, 45°C, and 50°C within the selected range.

In Fig. 7(a), it can be observed that the system's net work performance decreases as the gas cooler pressure increments by 200 kPa. This may occur because, as the gas cooler pressure is increased, the turbine work is reduced, and the LPC and HPC consumed work is increased. Consequently, when the gas cooler outlet temperature is maintained at 35°C, it demonstrates the highest net work value, while 50°C being the lowest. This could also be because higher outlet temperatures may lead to increased irreversibility or losses in the system. It is also noticed that, at gas cooler pressure 9000–9100 kPa, and outlet temperature 35°C, the flash-tank does not operate, thus shows system failure, which is true for the rest of the parameters as well. Similarly, if the gas cooler outlet temperature is set at 50°C and with the pressure range of 9000–9700 kPa, the system indicates a malfunction as the LTR and HTR does not operate under these conditions as the temperature differences are less than 5°C. Thus, these values have been excluded for all the graphs as well.

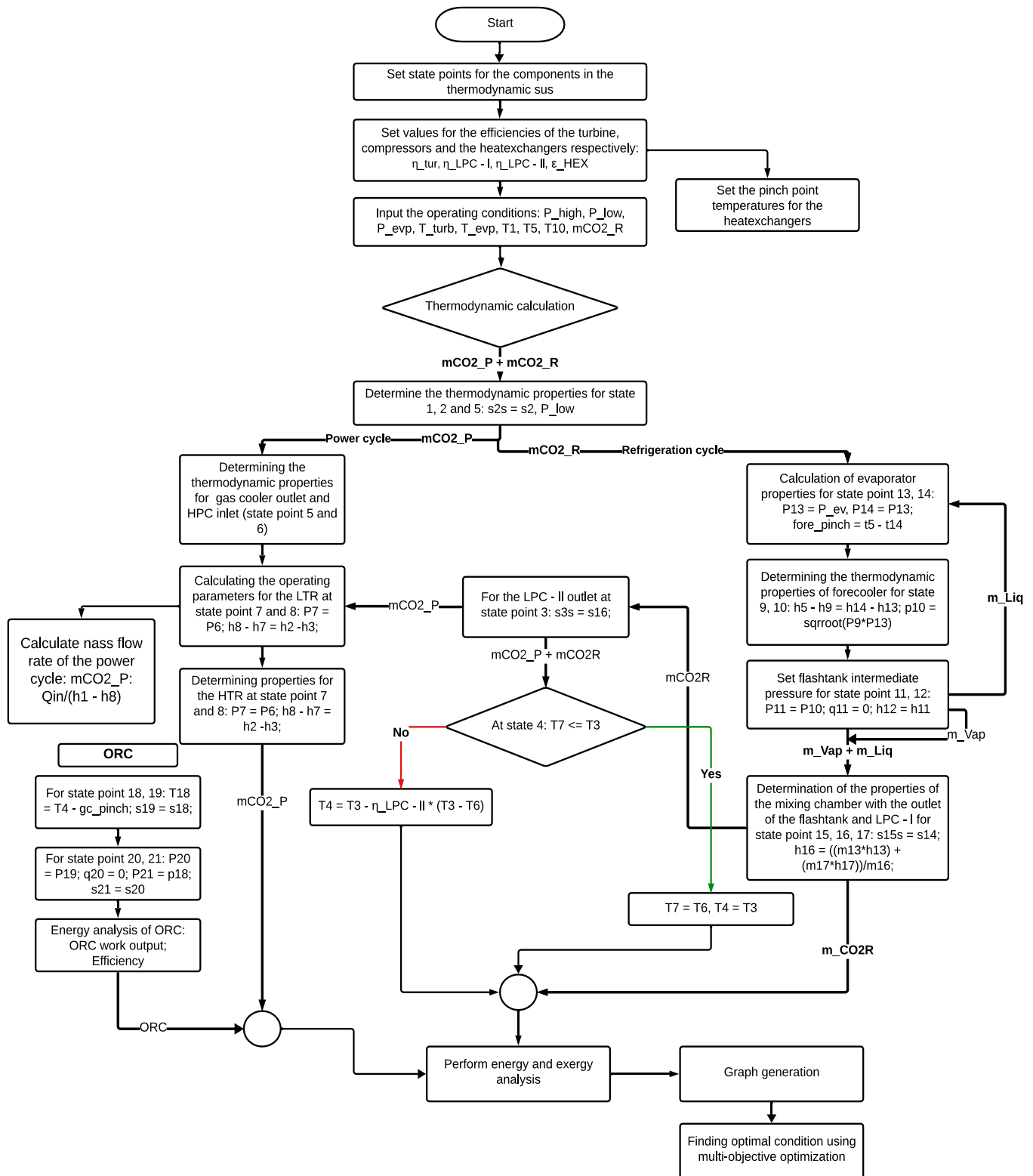


Fig. 5. Flowchart for the MERS system validation.

On the contrary, the net cooling of the refrigeration system increases as the gas cooler pressure increases, as shown in Fig. 7(b). As CO₂ comes closer to its critical temperature at 35°C, the most heat absorption happens at that temperature. Conversely, as the gas cooler outlet temperature increases, less heat is absorbed, hence 50°C shows the lowest net cooling effect. However, for all temperatures, the net cooling of the refrigeration system increases as the gas cooler pressure increases. This

could be because, a higher gas cooler pressure allows for more effective heat absorption in the evaporator, resulting in a greater cooling effect.

In Fig. 8, it is evident that, when the gas cooler outlet temperature is set at 35°C and 40°C, COP values tend to remain relatively constant at rising gas cooler pressure. This could be because the system is operating under conditions where the cooling effect in the evaporator and the work input that happens in the LPCs are both influenced by changes in

Table 5
Model validation table of reference combined power and refrigeration cycle: Energy analysis (A. Yu et al., 2020).

Parameters	Combined power and refrigeration cycle		Relative difference (%)
	Present Work	Ref (A. Yu et al., 2020)	
$\dot{Q}_{in}$ (kW)	14548.30	14600.91	0.40
$\dot{Q}_{out}$ (kW)	14515.18	14589.17	0.50
$\dot{Q}_{HTR}$ (kW)	23462.58	23404.32	0.20
$\dot{Q}_{LTR}$ (kW)	1518.18	1529.14	0.70
$\dot{Q}_{fore}$ (kW)	791.52	801.85	1.30
$\dot{W}_{tur}$ (kW)	5222.37	5223.14	0.01
$\dot{W}_{HPC}$ (kW)	975.09	975.24	0.02
$\dot{W}_{LPC}$ (kW)	1132.08	1145.46	1.20
$\dot{W}_{net}$ (kW)	3115.20	3102.44	0.40
$\dot{Q}_{ref}$ (kW)	3082.08	3090.70	0.30
$\eta_{P,1}$	29.19	29.09	0.30
COP	2.72	2.70	0.70
$\eta_{PR,1}$	42.60	42.42	0.40
$\eta_{WHR,1}$	40.57	40.71	0.30

Table 6
Model validation for the combined power and refrigeration cycle: Exergy analysis.

Parameters	Combined power and refrigeration cycle		Relative difference (%)
	Present Work	Ref (A. Yu et al., 2020)	
$\dot{E}_{in,PR}$ (kW)	8455.55	8479.85	0.30
$\dot{I}_{HRHE}$ (kW)	194.85	195.94	0.60
$\dot{I}_{HTR}$ (kW)	2770.31	2785.78	0.60
$\dot{I}_{LTR}$ (kW)	140.51	143.78	2.30
$\dot{I}_{cooler}$ (kW)	1085.20	1091.63	0.60
$\dot{I}_{fore}$ (kW)	51.86	53.14	2.40
$\dot{I}_{eva}$ (kW)	550.40	542.14	1.50
$\dot{I}_{tur}$ (kW)	243.27	243.42	0.06
$\dot{I}_{HPC}$ (kW)	134.46	134.57	0.08
$\dot{I}_{LPC}$ (kW)	182.27	184.17	1.03
$\dot{I}_{thrott}$ (kW)	210.29	211.63	0.60
$\dot{I}_{total}$ (kW)	5563.3	5586.20	0.40
$\eta_{2,PR}$ (%)	39.48	39.05	1.10
$\eta_{2,WHR}$ (%)	54.00	55.52	2.70

gas cooler pressure in a way that tends to balance out, resulting in a relatively constant COP. However, for gas cooler outlet temperatures of 45°C and 50°C, there is a tendency for COP values to rise with increasing gas cooler pressure. This indicates that higher gas cooler pressures are more beneficial for achieving better COP under these temperature conditions, which allows for a more effective cooling effect relative to the energy input required for compression.

Regarding the first and second law thermodynamic efficiencies shown in Fig. 9(a) and Fig. 9(b), both exhibits a similar pattern. Both efficiencies tend to decrease at rising gas cooler pressure. Specifically, at a gas cooler outlet temperature of 35°C, both efficiencies show the least values. The decrease in efficiency at 35°C indicates that higher gas cooler pressures are associated with reduced overall performance as less work is produced and the outlet temperature of the gas cooler is close to the CO₂ critical temperature, leading to an increased back pressure. On the contrary, at a gas cooler outlet temperature of 50°C, both efficiencies

Table 7
Thermodynamic state point properties of MERS.

State Point	P (MPa)	T (°C)	h (kJ kg ⁻¹)	s (kJ kg ⁻¹ K ⁻¹)	$\dot{m}$ (kg s ⁻¹)
1	20.00	550.00	1035.13	2.7411	64.36
2	10.00	466.44	942.03	2.7551	64.36
2 s	10.00	457.16	931.69	2.7411	64.36
3	10.00	79.44	469.79	1.8354	84.36
3 s	10.00	78.06	467.19	1.8280	84.36
4	10.00	54.54	406.01	1.6470	84.36
5	10.00	35.00	289.52	1.2806	20.00
6	20.00	51.78	304.37	1.2851	64.36
6 s	20.00	51.76	302.88	1.2806	64.36
7	20.00	58.56	320.91	1.3355	64.36
8	20.00	353.18	793.15	2.4052	64.36
9	10.00	27.55	263.91	1.1965	20.00
10	5.66	19.46	263.91	1.2156	20.00
11	5.66	19.46	254.05	1.1818	18.73
12	3.20	-3.19	254.05	1.2015	18.73
13	3.20	5.45	446.16	1.9124	18.73
14	3.20	25.00	471.77	2.0013	18.73
15	5.66	74.31	505.96	2.0195	18.73
15 s	5.66	69.38	499.67	2.0013	18.73
16	5.66	32.51	443.80	1.8280	20.00
17	5.66	19.46	408.98	1.7113	1.27
18	3.50	49.54	322.72	1.3925	103.69
19	0.146	15.13	312.34	1.3925	103.69
20	0.146	9.52	222.27	1.0799	103.69
21	3.50	10.78	227.93	1.0799	103.69

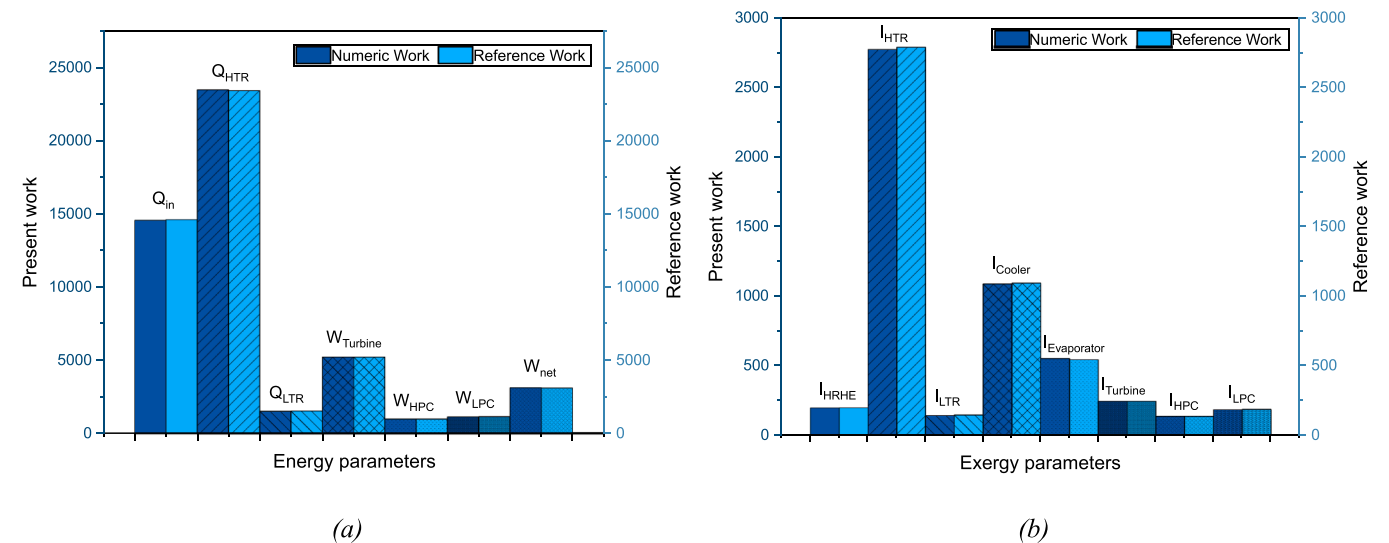


Fig. 6. Validation of thermodynamic system: (a) Energy parameters; (b) Exergy parameters.

Table 8
Comparison between MERS and reference system: According to 1st law of thermodynamics.

Energy Analysis		
Energy parameters	Reference System	Proposed System
$\dot{Q}_{in}$ (kW)	14548.30	15575.14
$\dot{Q}_{out}$ (kW)	14515.18	9828.08
$\dot{Q}_{HTR}$ (kW)	23462.58	30395.07
$\dot{Q}_{LTR}$ (kW)	1518.18	1064.74
$\dot{Q}_{fore}$ (kW)	791.52	512.20
$\dot{W}_{tur}$ (kW)	5222.37	5992.44
$\dot{W}_{HPC}$ (kW)	975.09	955.93
$\dot{W}_{LPCi}$ (kW)	-	683.81
$\dot{W}_{LPC }$ (kW)	-	519.95
$\dot{W}_{LPC}$ (kW)	1132.08	1203.76
$\dot{W}_{net}$ (kW)	3115.20	3832.75
$\dot{Q}_{ref}$ (kW)	3082.08	3597.78
$\eta_{P,1}$ (%)	29.19	39.25
COP	2.72	2.99
$\eta_{PR,1}$	42.60	54.62
$\eta_{WHR,1}$	40.57	41.17
$\dot{W}_{in,ORC}$ (kW)	-	587.40
$\dot{W}_{out,ORC}$ (kW)	-	1076.72
$\dot{W}_{net,ORC}$ (kW)	-	1076.13
η_{ORC} (%)	-	10.95

Table 9
Comparison between MERS and reference system: according to 2nd law of thermodynamics.

Exergy Analysis		
Exergy parameters	Reference System	Proposed System
$\dot{E}_{in,PR}$ (kW)	8455.55	9227.30
$\dot{I}_{HRHE}$ (kW)	194.85	313.87
$\dot{I}_{HTR}$ (kW)	2770.31	2974.07
$\dot{I}_{LTR}$ (kW)	140.51	415.96
$\dot{I}_{cooler}$ (kW)	1085.20	302.77
$\dot{I}_{fore}$ (kW)	51.86	27.60
$\dot{I}_{eva}$ (kW)	550.40	695.99
$\dot{I}_{tur}$ (kW)	243.27	279.28
$\dot{I}_{HPC}$ (kW)	134.46	90.74
$\dot{I}_{LPCi}$ (kW)	-	148.74
$\dot{I}_{LPC }$ (kW)	-	45.53
$\dot{I}_{LPC}$ (kW)	182.27	194.27
$\dot{I}_{thrott}$ (kW)	210.29	280.85
$\dot{I}_{flash}$ (kW)	-	221.89×10^{-10}
$\dot{I}_{mixch}$ (kW)	-	60.62
$\dot{I}_{total}$ (kW)	5563.3	5586.01
$\eta_{PR,2}$ (%)	39.48	54.90
$\eta_{WHR,2}$ (%)	54.00	54.54

exhibit higher values as that operate in a much more favourable temperature range.

The first and second law waste heat recovery efficiencies display a similar trend in Fig. 10(a) and Fig. 10(b). At a temperature of 35°C in the gas cooler outlet, these efficiencies remain constant with increasing gas cooler pressure, giving no significant improvement in waste heat recovery. When the gas cooler outlet temperature is set at 40°C, there is a slight increase in efficiency which quickly levels off after a gas cooler pressure of 9600 kPa. However, for gas cooler outlet temperatures of 45°C and 50°C, the efficiencies increase as the pressure rises, suggesting that they may approach a plateau or remain relatively constant with further pressure increments. This indicated that higher pressures are somewhat more favorable for improving waste heat recovery.

Fig. 11 presents the impact of operational parameters such as gas cooler pressure and evaporator pressure changes on the net work, COP,

and the 1st and 2nd law efficiencies of the MERS. As shown in Fig. 11(a), a notable improvement in the net work is evident when the evaporator pressure increases while maintaining a constant gas cooler pressure, which can happen as higher evaporator pressure allows for more heat to be absorbed from the surroundings, leading to a greater net work output. Conversely, rising gas cooler pressure decreases the overall net work. Furthermore, Fig. 11(b) demonstrates that an increase in the evaporator pressure leads to a relatively higher increase in COP, which holds true for an increase in gas cooler pressure as well, albeit at a much smaller scale. The reason behind it would be, as the evaporator pressure increases heat can be transferred more efficiently, resulting in higher COP.

Similarly, Fig. 12(a) and Fig. 12(b) exhibit comparable trends for the efficiencies, with both efficiencies increasing as the evaporator pressure rises but experiencing a slight decline when the gas cooler pressure is increased. The 1st law efficiency increases as, higher evaporator pressure leads to more efficient utilization of energy, thus the net cooling capacity increasing and so the system converts its input energy into useful work more effectively increasing its 2nd law efficiency as well. As for the gas cooler pressure, higher gas cooler pressure might increase the back pressure on the system, making it less efficient overall. It could also indicate that the benefits of increasing the evaporator pressure may be offset by an increase in gas cooler pressure in MERS.

6.4. Impacts of evaporator temperature

The evaporator temperature affects the heat absorption capacity of the system and its overall efficiency. By analysing different temperature of the evaporator, we can determine the changes in this parameter influence the net work output, COP, and thermodynamic efficiencies. Figs. 13–15 is provided to show how the system’s parameters are influenced by variations in gas cooler pressure, ranging from 9000 kPa to 10,000 kPa across four distinct evaporator temperatures, notably 5°C, 10°C, 15°C, and 20°C. In Fig. 13(a), it can be clearly seen that the system’s net work experiences a significant decline at increasing gas cooler pressure. Particularly, the net work reaches its maximum value at 5°C and gradually decreases as the evaporator temperature increases, ultimately reaching its minimum at 20°C. This trend may be attributed to increased heat absorption requirements or reduced efficiency in heat transfer processes. Moreover, it is observed that decreasing the evaporator temperature results in an increase in output power. Specifically, a 5°C increment in evaporator temperature results in the reduction of net work by about ~23 kW. This may be due to the substantial increase in LPC energy consumption with increasing evaporator temperature. Additionally, as evaporator temperature rises, the enthalpy difference during evaporation decreases, leading to a lower refrigeration capacity obtained at higher evaporation temperatures. The COP in Fig. 13(b) also follows a decreasing trend as gas cooler pressure increases, although this decrease is not as rapid. Evidently, the highest COP is observed at an evaporator temperature of 5°C, while the lowest COP is associated with a temperature of 20°C. For COP, a 5°C increment in evaporator temperature results in the COP decreasing by about 0.09. Higher gas cooler pressure and higher evaporator temperatures tend to reduce the COP due to increased work input required for the compressor.

The energy and exergy efficiencies shown in Fig. 14(a) and Fig. 14(b) exhibit a consistent decreasing pattern, mirroring the previous observations. Specifically, the highest efficiencies are achieved at an evaporator temperature of 5°C, while the lowest efficiencies occur at 20°C. Energy efficiency ranges approximately from 51 % to 61 %, with a difference of about 1.23 % with every 5°C increment in evaporator temperature. In contrast, exergy efficiency falls within a steeper range of 53–67 %, showing a difference of approximately 1 % with each 5°C increment in evaporator temperature. These trends indicate that the system’s ability to convert input energy into useful work (both in terms of energy and exergy) decreases as the pressure of the gas cooler and evaporator temperature increase. On the contrary, the waste heat recovery efficiency for both the first and second law tend to increase as the

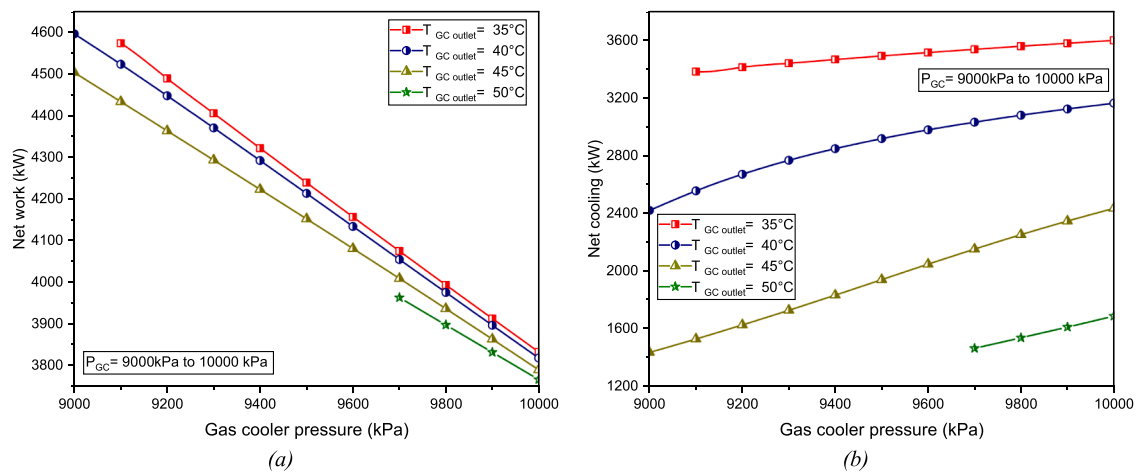


Fig. 7. Impact of gas cooler pressure at varying gas cooler outlet temperatures on the (a) net work and (b) net cooling.

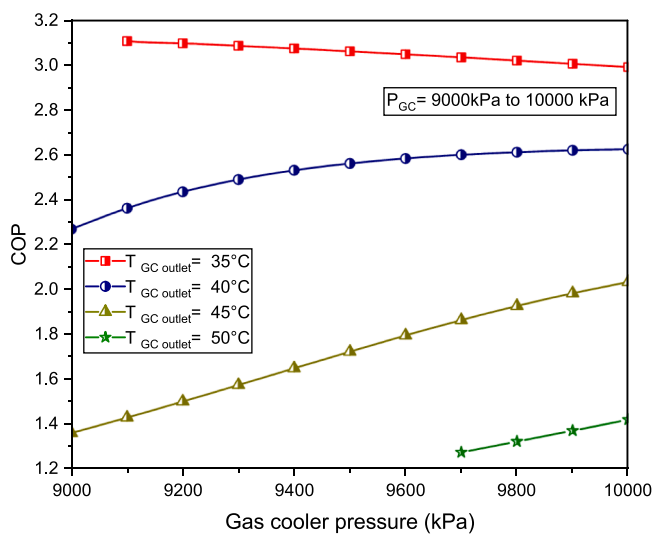


Fig. 8. Impact of gas cooler pressure at varying gas cooler outlet temperatures on the COP.

gas cooler pressure increase, but lowest values can be found as the evaporator temperature decreases, which is represented by Fig. 15(a) and Fig. 15(b). So, it can be said that, increasing gas cooler pressure can improve the system's ability to recover waste heat, but low evaporator temperatures may limit this recovery efficiency.

6.5. Impacts of turbine inlet temperature

The turbine inlet temperature is a critical parameter that impacts the power output and efficiency of the system and gives the idea of the potential improvement of the system performance under certain conditions. Fig. 16 and Fig. 17 provide insights into how turbine inlet temperature affects the system's parameters. In Fig. 16(a), it was observed that, the net work decreases as the turbine temperature increases, ranging from 350°C to 550°C with 50°C increments. Even though the power consumption by compression remains constant under the pre-set conditions, the overall turbine power generation increases as the enthalpy difference increases with rising temperature at the turbine inlet (A. Yu et al., 2020). However, this increase is less than the decrease in net work due to a drop in mass flow rate (Fig. 18). Therefore, when the temperature of the turbine inlet rises, the power cycle's net work always drops. Furthermore, it is evident that as the outlet temperature of the gas cooler increases, the decline in net work becomes more

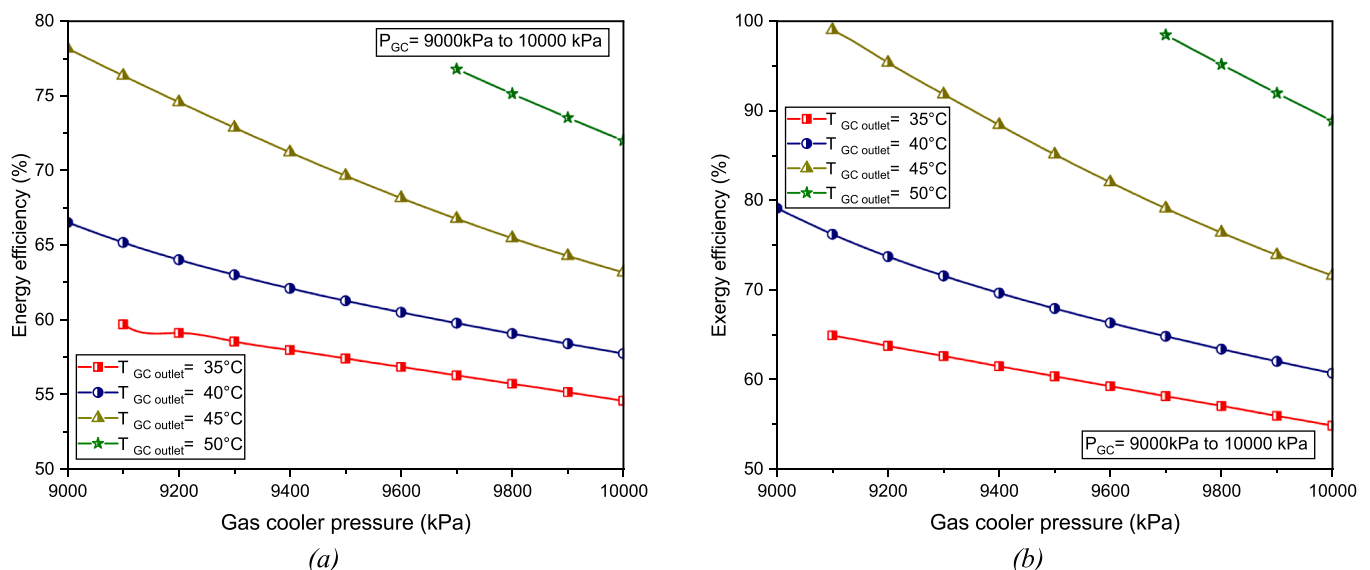


Fig. 9. Impact of gas cooler pressure at varying gas cooler outlet temperatures on the (a) Energy and (b) Exergy efficiency.

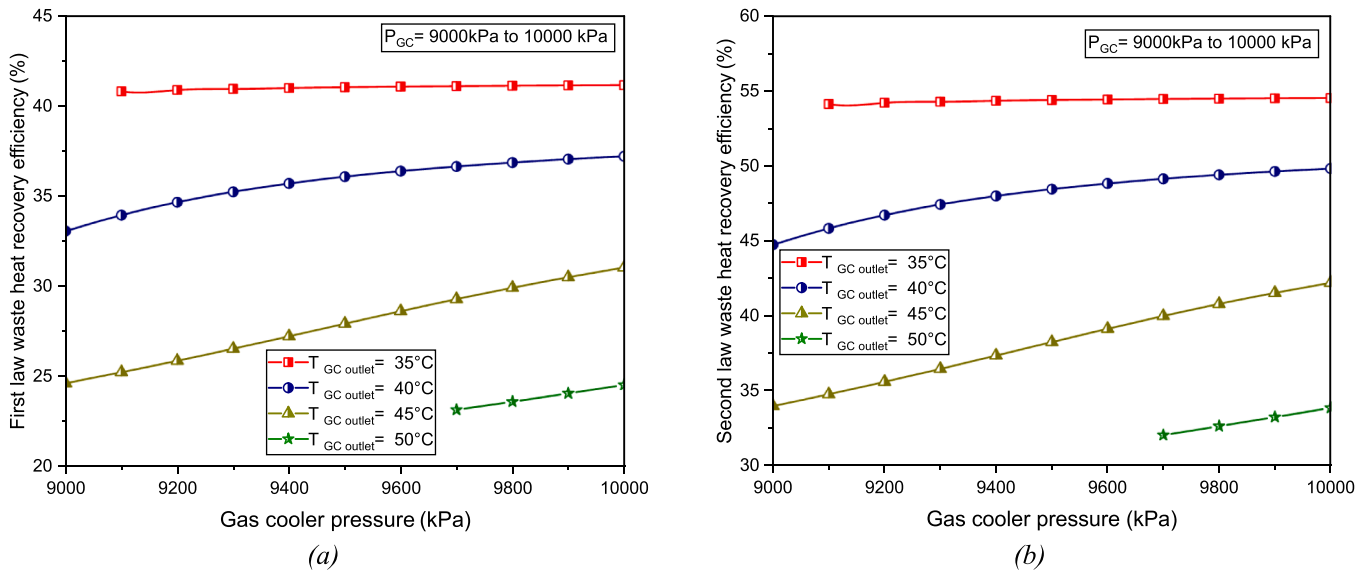


Fig. 10. Impact of gas cooler pressure on waste heat recovery efficiency at varying gas cooler outlet temperatures in terms of the (a) First law, and the (b) Second law.

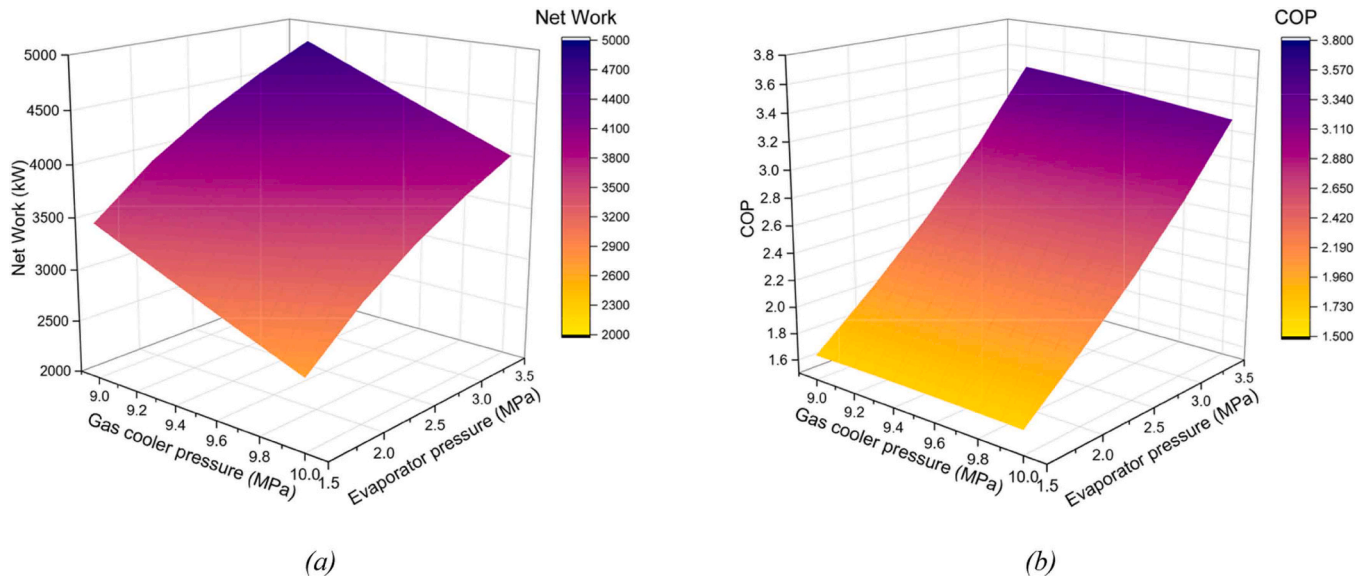


Fig. 11. Impact of evaporator and gas cooler pressure on MERS's (a) net work and (b) COP.

pronounced, indicating reduced net work from the system. Conversely, when the gas cooler outlet temperature is kept lower, the deviation in net work with an increase in turbine inlet temperature is less pronounced. In Fig. 16(b) and Fig. 16(c), it can be observed that the trends in the first law and second law efficiencies that the energy efficiency increases with the rise in turbine inlet temperature, whereas exergy efficiency increases slightly before declining. However, for both efficiency metrics, a higher temperature at the gas cooler outlet is advantageous.

Regarding waste heat recovery, both Fig. 17(a) and Fig. 17(b) illustrates a similar trend of rapid decline as the turbine inlet temperature increases. In this scenario, lower gas cooler outlet temperatures result in higher efficiency values. This indicates that as the turbine inlet temperature increases, it becomes more challenging to recover waste heat efficiently, and maintaining a lower gas cooler outlet temperature helps mitigate this decrease in recovery efficiency. With rising turbine inlet temperature, Fig. 18 illustrates a decrease in net work output. Additionally, the mass flow rate exhibits an almost linear decrease in

response to the rising temperature. In contrast, the specific work of the cycle initially increases, reaching a peak value, after which it starts to decline slightly. The decrease in net work output with increasing turbine inlet temperature might have happened due to evident decreasing mass flow rate.

6.6. Impacts of isentropic efficiencies

In this section, the isentropic efficiencies of the HRHE, turbine, HPC, and LPCs are maintained at fixed values of 90 % for all the components. However, it's crucial to recognize that isentropic efficiency is intricately connected to the system's operating conditions. As these conditions change, alterations in turbomachinery efficiency can exert a substantial influence on the overall system performance, as shown in reference (A. Yu et al., 2020). Therefore, this study encompasses a range of turbomachinery efficiencies to comprehensively assess their impact on the MERS performance. Shown in Fig. 19(a), when the efficiency of the

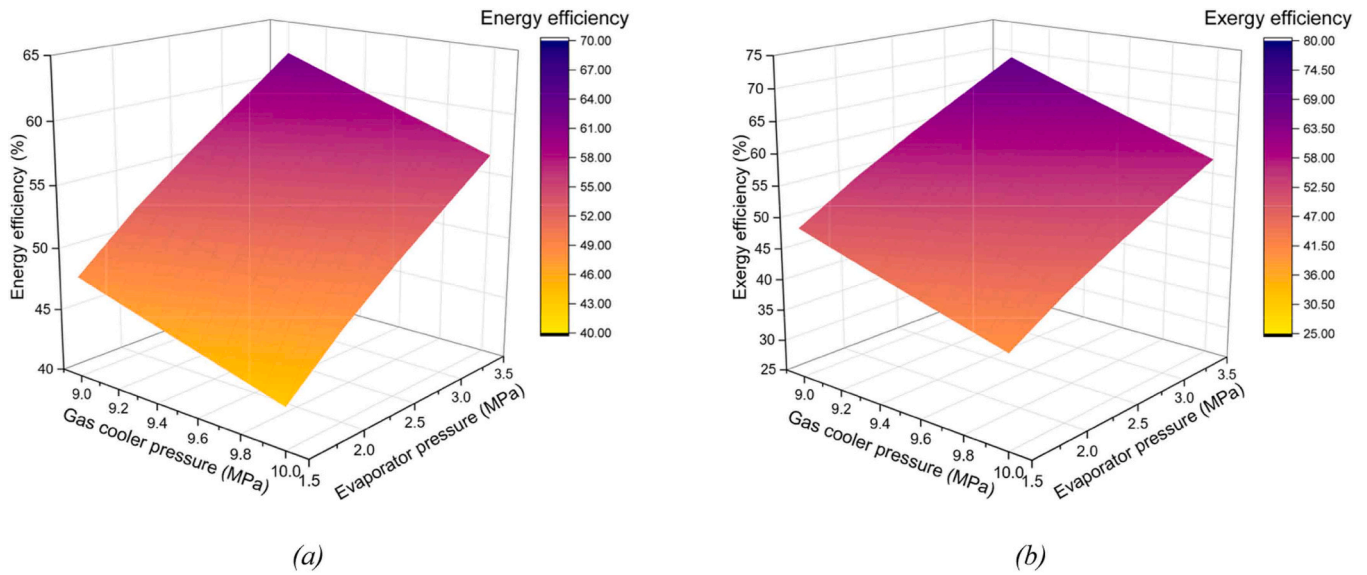


Fig. 12. Impact of evaporator and gas cooler pressure on MERS's (a) energy and, (b) exergy efficiency.

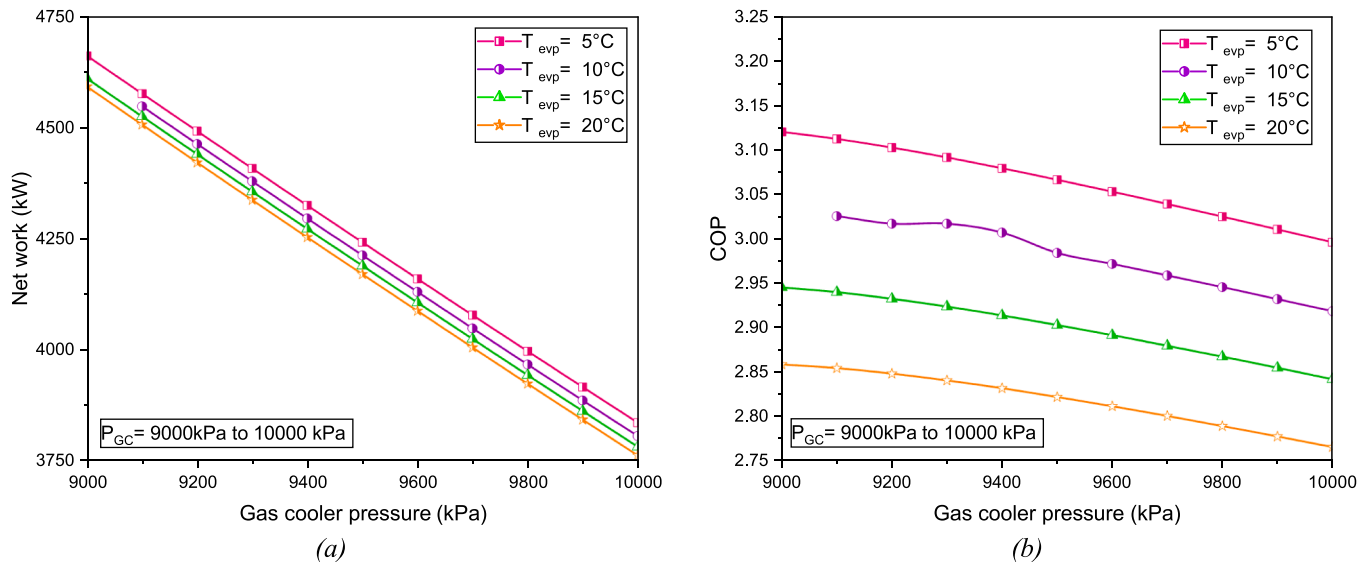


Fig. 13. Impact of gas cooler pressure at varying evaporator outlet temperatures on the (a) net work and (b) COP of the MERS.

HRHE is increased, the net work grows from 3373 kW to 3881 kW. Similar trend is also reflected in the energy and exergy efficiency of the system, which slightly increase from around 40.19–41.27 % and 53.40–53.66 %, respectively. This trend is likely because as the HRHE becomes more efficient, it transfers more heat from one side to the other with less loss. This means that more heat is available for the total work done in the system, resulting in an increase in net work and efficiencies. However, the increase in energy and exergy efficiency is not significant with the added cost for acquiring a HRHE of higher efficiency. Moving on to Fig. 19(b) and considering the isentropic efficiency of the turbine, increasing turbine efficiency leads to an increase in network, energy efficiency, and exergy efficiency. The net work linearly rises from approximately 1899 kW to 4149 kW with higher turbine efficiency. This increase is attributed to the enlarged enthalpy difference between the turbine's inlet and outlet, resulting in greater work output. Furthermore, as turbine efficiency increases, both energy and exergy efficiencies increase, with energy efficiency going from 36.63 % to 41.92 % and exergy efficiency from 49.13 % to 55.42 %.

In Fig. 20(a), the isentropic efficiency of the HPC is varied. As the

efficiency of the HPC increases - the net work, energy and, exergy efficiency increase, going from approximately 3373 kW, 40.19 %, and 53.39 % to 3881 kW, 41.27 %, and 54.66 %, respectively, which shows that compared to turbine efficiency, the isentropic efficiency of HPC has a relatively smaller impact on system performance. As the difference of enthalpy in the HPC decreases, the work done by HPC decreases, thus making the net work increase almost linearly. And so, as the energy and exergy efficiency increase. Similarly, Fig. 20(b) explores the effects of the LPC efficiency on cycle performance. As LPC efficiency increases, performance indicators such as energy efficiency, and exergy efficiency, decreases except the net work. Specifically, with higher LPC efficiency, the net work increases from about 2942 kW to 3930 kW, whereas the energy efficiency, and exergy efficiency decrease 42.86–56.52 %, and 40.99–54.33 %, respectively. As LPC efficiency increases, the exergy destruction within the system decreases because it operates more efficiently, however, the overall system's exergy efficiency might decrease due to the redistribution of the exergy destruction through other components. This analysis underscores that the development of combined system for waste heat recovery should always consider acquiring a

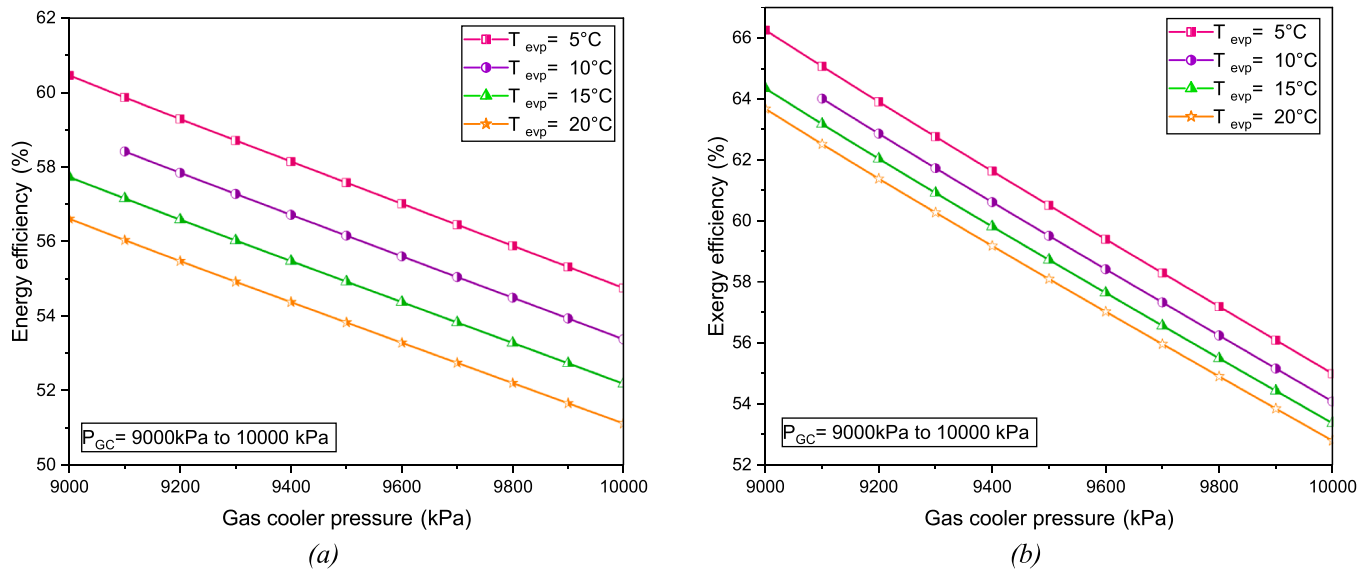


Fig. 14. Impact of gas cooler pressure at varying evaporator temperatures on (a) energy and (b) exergy efficiency of MERS.

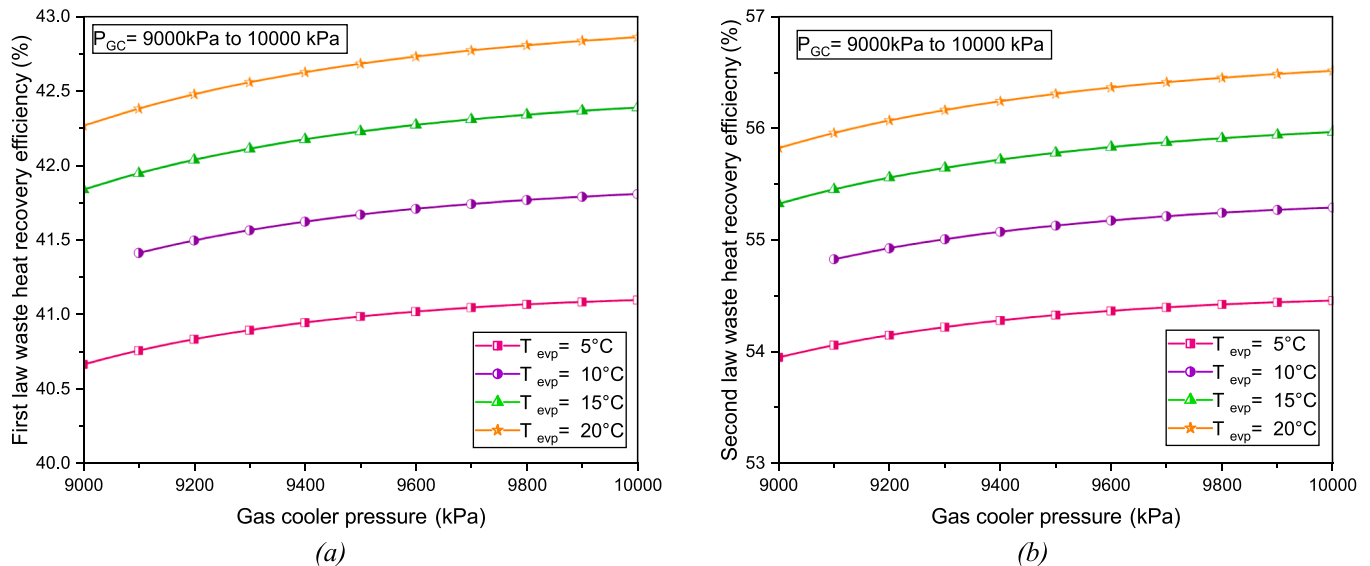


Fig. 15. Impact of gas cooler pressure at varying evaporator outlet temperatures on the (a) first law thermodynamic waste heat recovery efficiency and, the (b) second law waste heat recovery efficiency.

higher efficiency turbine to have a greater return in investment compared to the HRHE and compressors.

6.7. Exergy destructions of the system

Analyzing exergy destruction in each component of the system helps in identifying where the most significant losses occur. With the gas cooler's outlet temperature maintained at 40°C , Fig. 21 shows the exergy destruction rate for each system component under various parameters. In Fig. 21(a), it becomes evident that, overall exergy destruction rate increases with rising gas cooler pressure, especially in the HTR, the evaporator, LTR, HRHE, LPCs and the turbine, in a descending order. As the gas cooler pressure increases, the exergy destruction rate of the HTR and LTR experiences a significant rise, while there is a slight decrease in irreversibility for the turbine, gas cooler, HPC, and LPC-I. This might be due to the fact that HTR and LTR works less efficiently due to less pressure difference within the cycle. Moving to Fig. 21(b), an opposite trend is observed. There is a decreasing trend

in irreversibility for HRHE, HTR, LPC-I, LPC-II, the evaporator, and the mixing chamber. However, the exergy destruction in the turbine stays almost constant as the evaporator pressure rises and results in less efficient heat exchanging in the evaporator. In Fig. 21(c), the exergy destruction increases in HTR but decreases in HRHE, turbine, LTR, gas cooler, and HPC with an increase in turbine inlet temperature. Other components maintain relatively constant exergy destruction values. It's worth mentioning that the exergy destruction in the flash-tank is negligible in comparison to the other components and is therefore not considered in this analysis.

6.8. Impacts of ORC turbine inlet pressure

Common ORC working fluids like R245fa, R123, and R152a has a cycle temperature lesser than 350°C to ensure the safer use of ORC systems (Dai et al., 2019). Different working fluids respond differently to changes in turbine inlet pressure, impacting energy and exergy efficiencies. Studying this parameter helps in selecting the most suitable

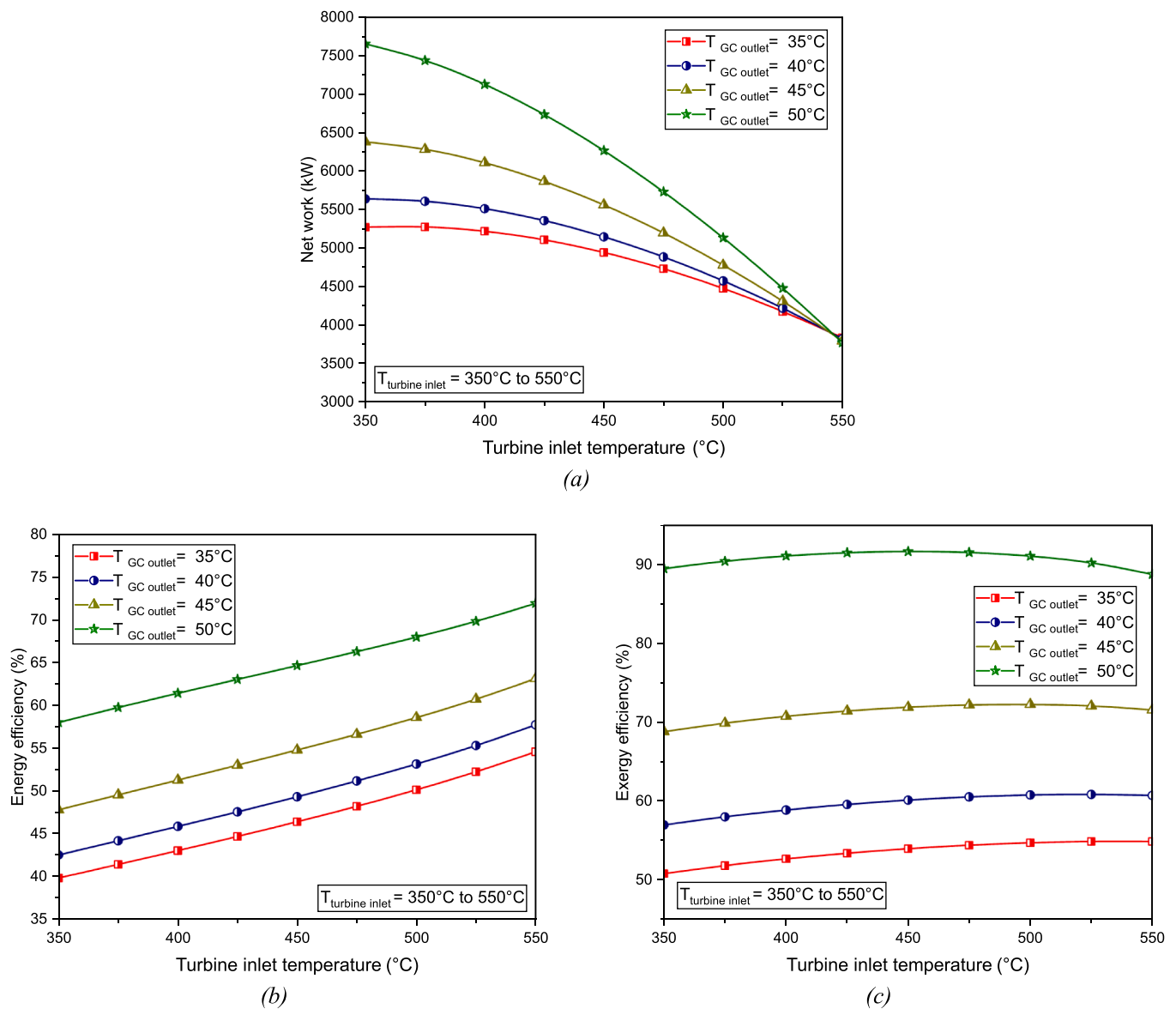


Fig. 16. Impact of turbine inlet temperature at varying gas cooler outlet temperatures on the (a) net work, (b) energy efficiency, and (c) exergy efficiency of the MERS.

working fluid and optimizing the ORC for maximum efficiency and power output. In this study, Butane is utilized as the primary working fluid for the ORC. However, an examination of a few other fluids for the system has also been conducted, as depicted in Fig. 22(a-b). Specifically, a parametric analysis has been conducted with Butane, R245fa, Toluene, and R123, as they are suitable for the conditions of our system. It was observed that the energy and exergy efficiencies of the system follow a similar trend across different working fluids as the turbine inlet temperature of the ORC increases. However, there are variations in the specific values for each fluid. For Toluene, energy efficiency ranges from about 45.93–46.29 % with an increment of 0.16 % on average, while exergy efficiency stays at 40.71 % constantly. It can be said that Toluene has moderate efficiency but relatively stable performance over the inlet temperature range of the turbine considered. For Butane, these values give a slightly decreasing trend, varying from approximately 54.82–54.60 % and 55.25–54.88 % for energy and exergy efficiency respectively. This indicates that Butane's efficiency starts relatively high but decreases as turbine inlet temperature rises. It might reach its upper limit for efficient operation within the considered temperature range. R245fa also shows a very slight decreasing trend as the turbine outlet

temperature increases, with minimal deviation in both energy and exergy efficiency. It shows that R245fa has energy efficiency changing from 53.89 % to 54.49 % and exergy efficiency being 53.67–54.69 % which behaves consistently with a increase in efficiency as the turbine inlet temperature rises, but the change in energy efficiency is relatively greater compared to Butane but the exergy efficiency is lesser. For the working fluid R123, efficiency increases with pressure, resulting in energy efficiency ranging from about 56.21–65.63 % and exergy efficiency from about 57.60–73.49 %, respectively, which shows a positive correlation between pressure and efficiency.

Compared to the reference system by Yu *et al.* the new system, MERS, demonstrates superior performance over the reference system, with net work output being 3832.75 kW from 3115.20 kW net cooling capacity being in a rising trend from 3082.08 kW to 3597.78 kW. Even though the parameters considered for this system is not the same, a significant improvement can still be noticed in the COP as well along with the exergy efficiency rising from 39.48 % to 54.90 %, and energy efficiency from 42.60 % to 54.62 %. These enhancements highlight the system's advanced capability in optimizing waste heat recovery and overall operational efficiency.

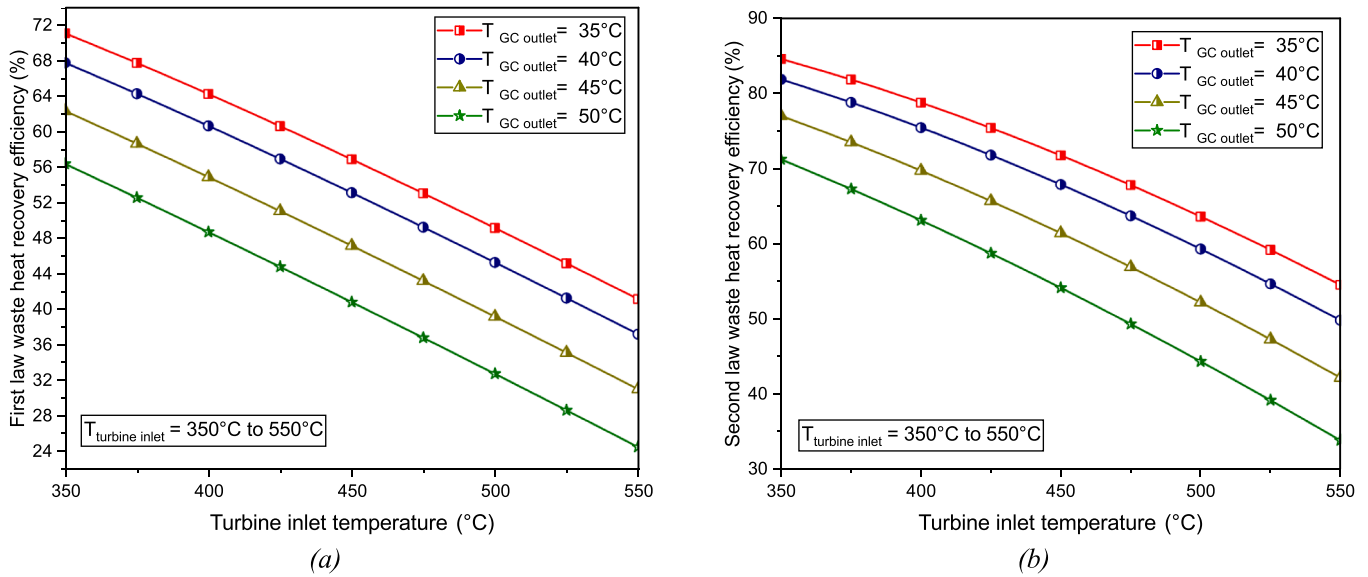


Fig. 17. Impact of turbine inlet temperature for varying gas cooler outlet temperatures on the (a) first law thermodynamic waste heat recovery efficiency and, the (b) second law waste heat recovery efficiency.

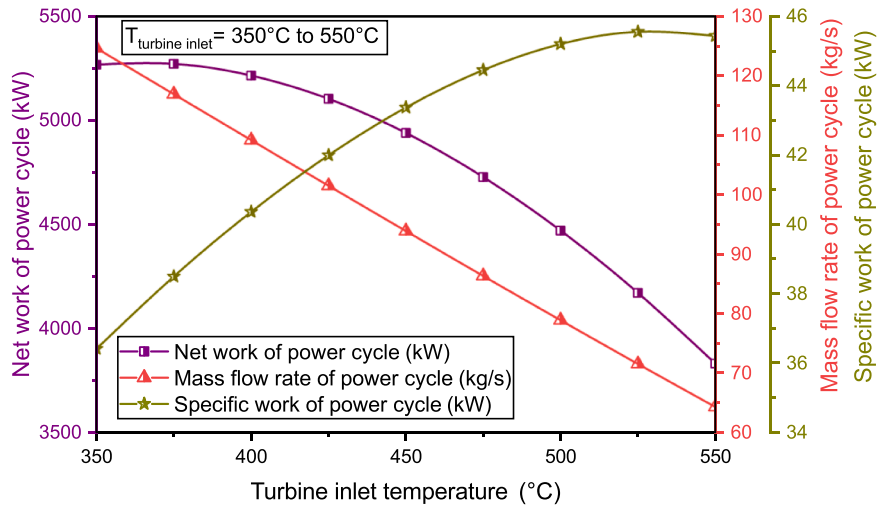


Fig. 18. Impact of temperature at the turbine inlet on the MERS's net work output, mass flow rate, and specific work of the power cycle.

7. Multi-objective optimization

7.1. Artificial Neural Network (ANN)

Artificial neural networks (ANNs) are computational systems that draw inspiration from the structure and learning behaviours of biological neuron cells. They serve as robust classifiers in the realm of pattern recognition. With the advancement of computer technology, ANNs have diverse applications such as: pattern recognition, system modelling, classification etc (Işcan, 2020; Karthikeyan et al., 2021). In summary, the ANN be described as the following:

- A mathematical model rooted in the functioning of biological neurons.
- Comprised of multiple processing elements interconnected.
- Information storage involves connection weights.
- Every processing element can react dynamically to input signals; however, the processing element's response depends solely on the local information, which includes connections and the effect of link weights on the particular processing element.

- It is able to learn, remember, and apply knowledge because of the link weights that are created during training with data.

The above characteristics make ANNs well-suited for solving complex problems.

7.2. Genetic Algorithm (GA)

Similar to ANN, which tries to emulate the structure and function of a neuronal cell to make predictions, Genetic Algorithm (or GA), is rooted in the concept of natural evolution, where stronger individuals have increased chances of passing on their genes (solutions) to offspring, ultimately leading to their dominance within the population and can employed to tackle both unconstrained and constrained optimization problems.

When comparing GA to fundamental optimization tools, a notable distinction emerges. Basic optimization algorithms generate a single point per iteration, gradually converging to the desired optimal solution through a sequence of points. In contrast to generating a single point, genetic algorithms generate datasets (population) of potential

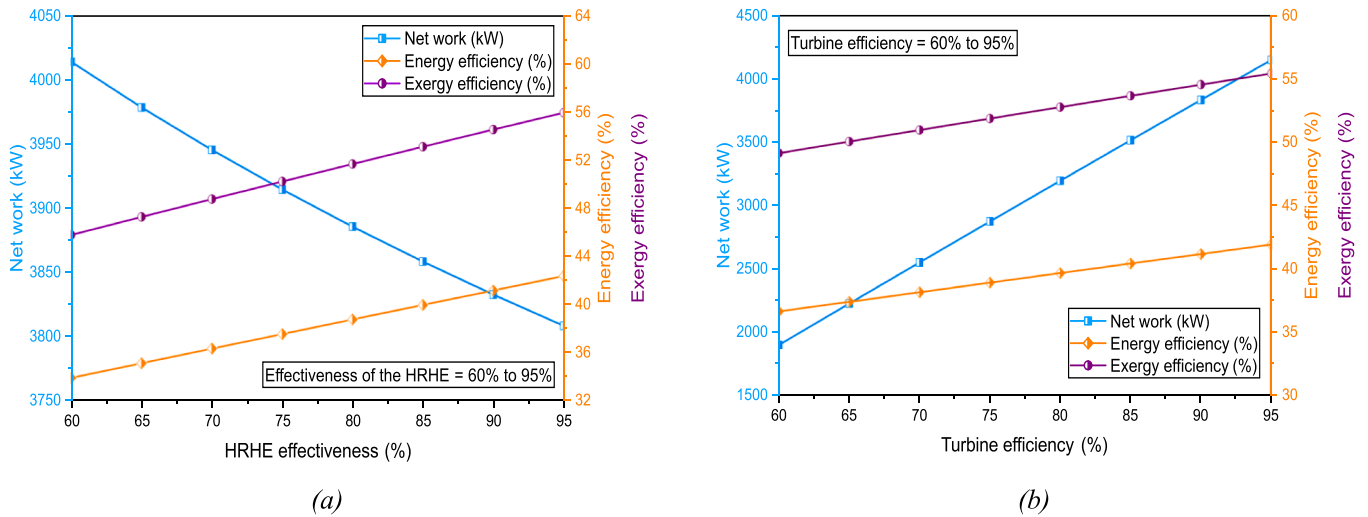


Fig. 19. Impact of the (a) heat exchanger and the (b) turbine on the system's net work, energy efficiency, and exertional efficiency.

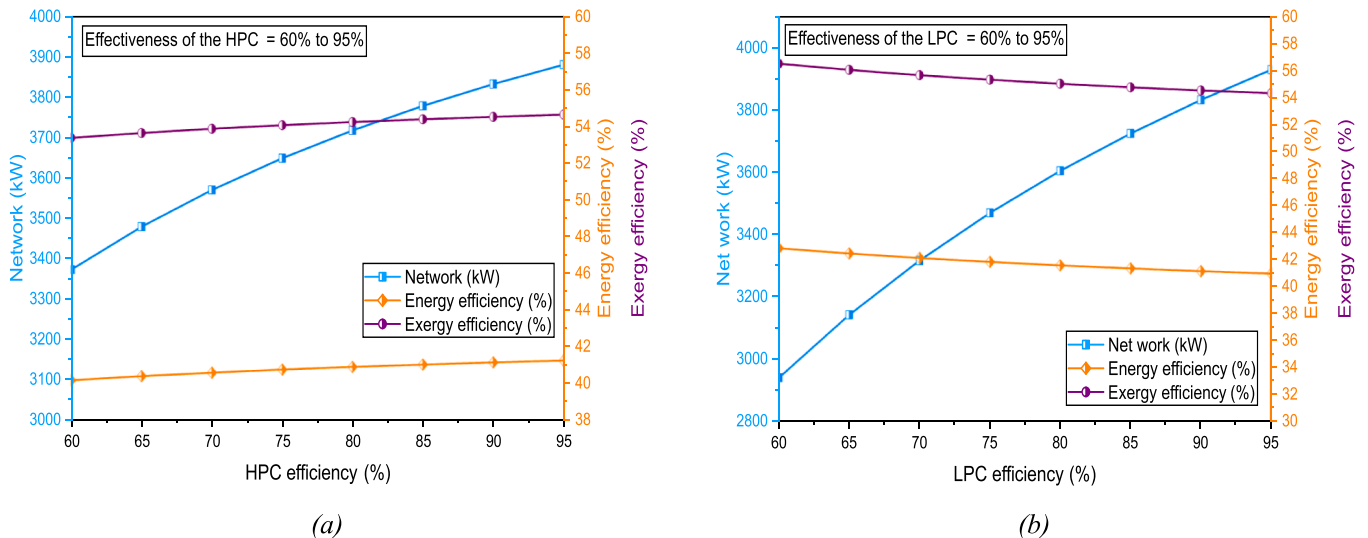


Fig. 20. Impacts of the (a) HPC and (b) LPC efficiency on the system's net work, energy, and exertional efficiency.

combinations, or solutions (coined as genes) at each iteration (Farsi et al., 2020). The algorithm involves iteratively modifying a collection of these individual solutions. During each step, the algorithm selects individuals to generate the subsequent iteration, continuing until an optimal solution is reached (Sohail, 2023). Within the framework of genetic algorithms, functions such as mutation and crossover are utilized to create new datasets at each iteration. Cross-overing involves generating new off springs by combining genetic material from parent individuals (Zhang et al., 2019). Parents are chosen from the current population based on fitness, with favourable values being the optimization parameters. Offspring inherit favourable genes from these competent parents. Through successive iterations, the genes of adept individuals become more prevalent across the population, ultimately driving convergence towards the optimal solution (Zhang et al., 2019). The convergence of the optimal solution can be due to many reasons, such as: the output values not having improvements above a certain threshold every iteration or reaching the limit of the number of iterations. Diversity at the genetic level is introduced through mutation, wherein the mutation rate governs changes in gene properties. This prevents the algorithm from getting stuck in local minima or maxima and aids in reaching the targeted solution.

7.3. Optimization methodology

The purpose of optimization is to improve the MERS to achieve optimal operating conditions. The maximum efficiency and net work performance are taken as output values to find the optimum values. The identification of decision variables that define the appropriate design options is a crucial component of an optimization problem. To analyse and assess the system, the proposed integrated cycle is simulated using Python software. After that, an evaluation of the system's reaction to changes in these input parameters and the target outputs, which will be supplied to the ANN for training, is done through a parameter sensitivity analysis. These include the pertinent elements that have an impact on the system performance. The turbine inlet temperature, gas cooler pressure and, evaporator pressure, are among the cycle's input variables. The gas cooler pressure serves as the lowest pressure point in the power cycle and the highest in the refrigeration cycle and its outlet temperature is also important for both the cycles. The turbine inlet temperature is a critical parameter in shipboard system as it is related to the exhaust gas temperature, and it directly impacts the thermal efficiency as well as the power output. In this study, the ANN model "2-layer Feed-Forward Back Propagation" algorithm is employed with the parameters used in Table 10. The back-propagation procedure is a very

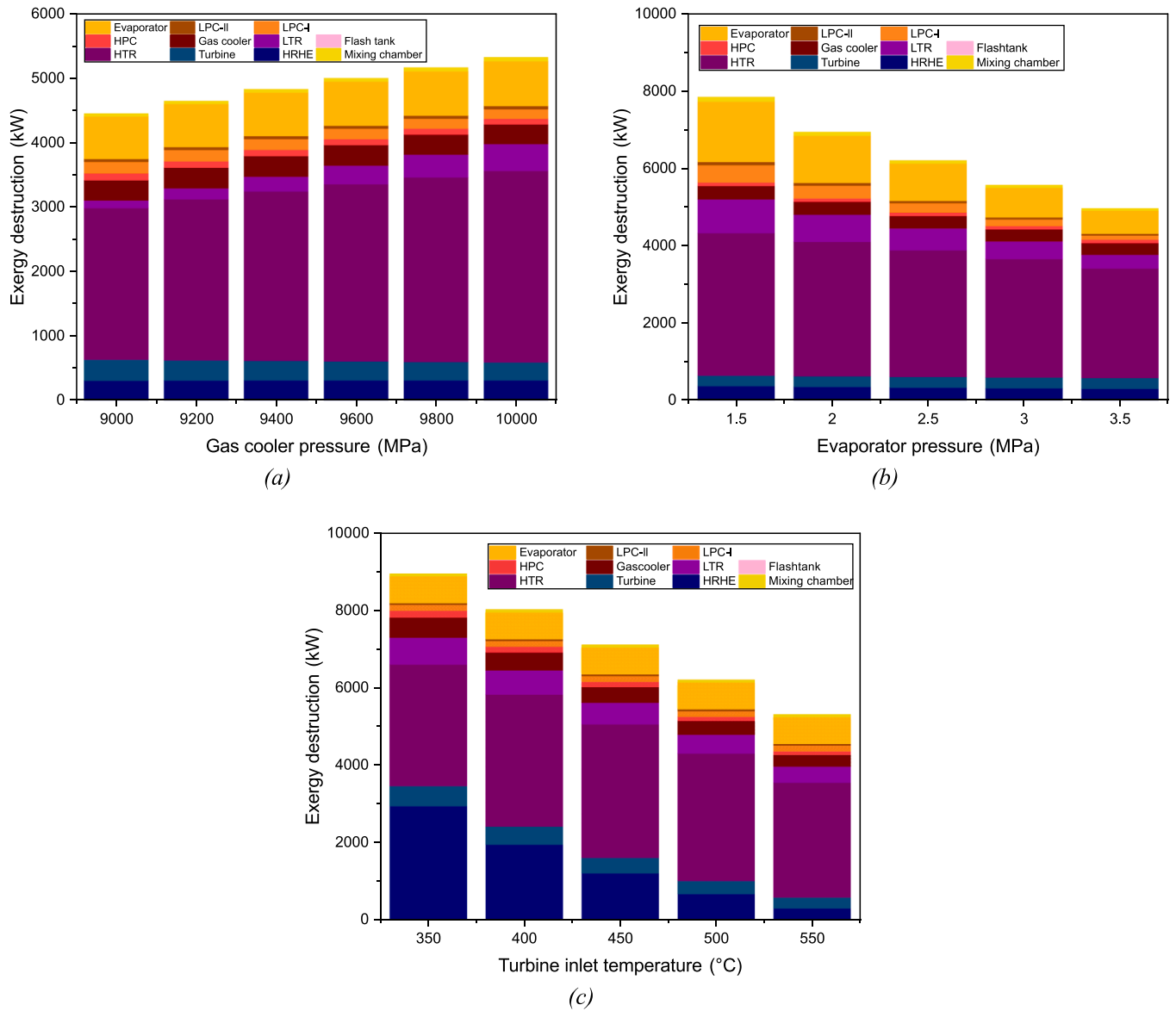


Fig. 21. Impacted of exergy destruction by (a) gas cooler pressure, (b) evaporator pressure, and (c) turbine inlet temperature of the MERS.

practical approach to calculate the gradient of an objective function concerning the weights within a multi-layer stack of modules. It enables the computation of the derivative of the objectives relative to the input module by retracing the steps from the gradient concerning the output of that module, effectively working backward through the network to compute the necessary gradients (LeCun et al., 2015). The cost function, mean squared error (MSE), and the coefficient of determination, R^2 is used. Eq. (48) is used to determine the MSE value:

$$MSE = \frac{\sum_{i=1}^N (t_i - p_i)^2}{N} \quad (48)$$

Where t_i is the predicted value and p_i is the actual value, and N is the number of data points.

To find the coefficient of determination, R^2 , the correlation coefficient between the prediction value and experimental data, or R value, is found to assess the performance of the ANN model for all pairs (Yang et al., 2018).

The R value is determined using Eq. (49) (Yang et al., 2018),

$$R = \frac{(t - \bar{t})(p - \bar{p})^T}{\sqrt{(t - \bar{t})(t - \bar{t})^T} \sqrt{(p - \bar{p})(p - \bar{p})^T}} \quad (49)$$

Here, the mean values of the experiment data and the prediction data are $\bar{t}$ and $\bar{p}$, respectively.

The sigmoid function is used in the hidden layer to provide non-linearity into the relationship between the data. The linear function is applied in the output layer to allow the model to handle regression tasks. Fig. 23 shows a neuronal diagram that represents the mathematical modelling of the ANN for this study.

Once the neural network setup is complete, the training process is initiated. The training method employed here is Bayesian Regulation, which enhances the accuracy of the system and prevent overfitting. The effectiveness of the ANN is strongly influenced by the quantity of hidden layers. Determining the optimal hidden layer quantity is crucial, as it results in the best alignment between the function created by the ANN and the inputs and outputs (Bicer et al., 2016). This determination is typically achieved through a trial-and-error process.

The process begins by generating the initial input dataset, followed by configuring the specification of input, output, and testing parameters.

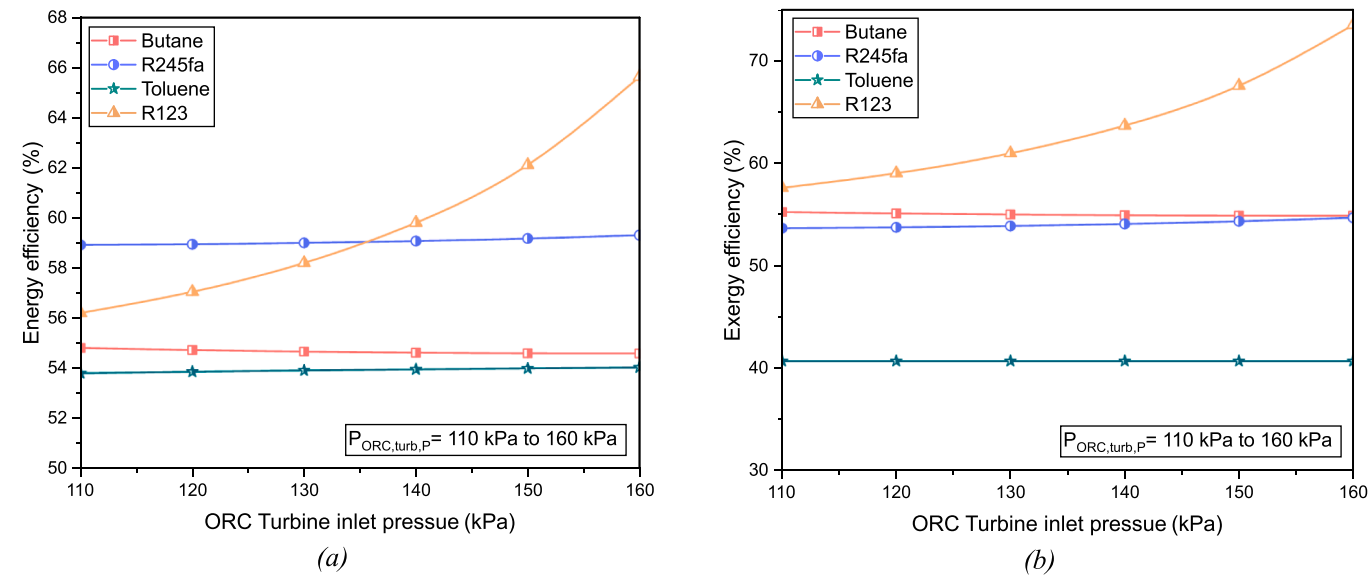


Fig. 22. Temperature at which the ORC turbine inlet opens affects the cycle’s (a) energy and (b) exergy efficiency for various working fluid types.

Table 10
Parameters used for the ANN model.

Parameters	Value/Range
Activation function (hidden layer)	Sigmoid
Activation function (output layer)	Pure linear
Training data used in percentage	70 %
Validation data used in percentage	16 %
Testing data used in percentage	14 %
Training method used in the optimization	Bayesian regularization
Hidden layer number	12
Simulating software used in the optimization	MATLAB

The phase involving the configuration of the neural network entails determining the appropriate number of neurons, selecting a performance function, and configuring hidden layers. If the MSE is below a certain error threshold, the subsequent steps are completed, which includes: preserving the trained neural network, capturing crucial data

such as MSE values, regression metrics, epochs, and raw data with accompanying graphs, and saving the trained ANN objective function (the function that will be optimized). In cases where the MSE is larger than the target threshold, the current network is terminated, and a new ANN architecture is devised. This initiates a loop back to the neural network configuration phase to make necessary adjustments and continue until the desired MSE goal is attained.

Following the training of the ANN model and acquiring the objective function, the optimization study is conducted to enhance the system’s performance using MATLAB. Genetic Algorithm is used to optimize the data to acquire a data of the most optimal outputs, using the parameters shown in Table 11. The convergence criterion was based on the tolerance of the average change between the spread in the solutions. Namely, the optimization would stop if the spread of the Pareto front solutions was less than the predefined tolerance threshold. The number of generations was 100 times the number of variables (default criterion), which prevented the optimization to be terminated due to reaching the

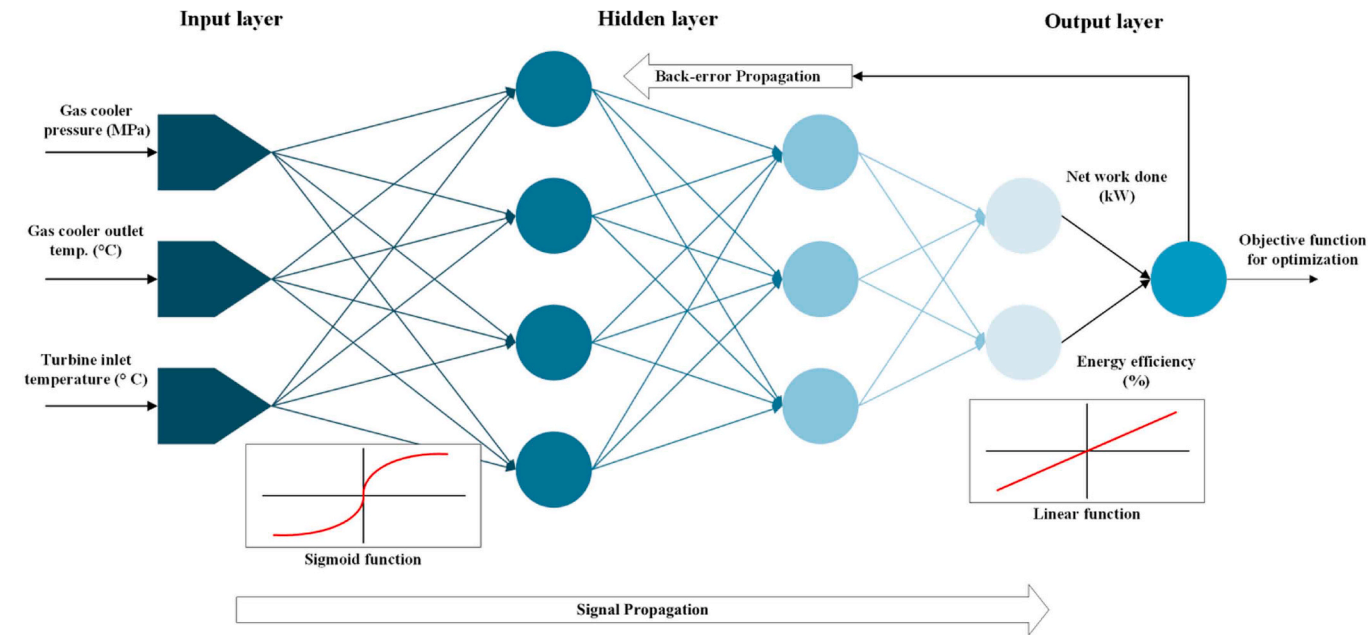


Fig. 23. Mathematical model of neuron cells.

Table 11
Parameters used for the Genetic Algorithm-based optimization.

Limits of input parameters		
Upper limits:	Gas cooler pressure: 10 MPa	
	Gas cooler outlet temperature: 50°C	
	Temperature in the inlet of the turbine: 550°C	
Lower limits:	Gas cooler pressure: 9.5 MPa	
	Gas cooler outlet temperature: 35°C	
	Temperature in the inlet of the turbine: 350°C	
Genetic Algorithm Parameters and their description		
Optimal principal	Pareto optimum points	
Simulating software	MATLAB	
Population size	200	Number of potential solutions per iteration. It is taken higher than the default value of 50 to improve its chances of finding the global optimum.
Function tolerance	0.0001	Tolerance level to accept the minimum change in the spread of output values.
Crossover function	singlepointcrossover	After selecting two “parent” solutions, part of its unique identifier or “genome” is swapped or “cross-overed” to produce two new identifiers (i.e., new potential solutions for next iteration)
Crossover fraction	0.8	The fraction of the population that undergoes crossover.
Migration fraction	0.2	The fraction of the population that are exchanged between “sub-populations”, which are random datasets made of part of the population.
Mutation function	mutationadaptfeasible	Mutation is the random changing of the genome of a solution to generate new solutions, that are otherwise impossible. This mutation function allows for the mutation to respect the constraints given by the user (upper bound, lower bound) and ensure the mutated solution remains within the constraints while also avoiding local optima.
Elite count	5 %	This is the fraction of the top-performing population that directly gets carried over to the next generation without any crossover or mutation.

iteration limit. The Pareto front method is used to identify the ideal points. The ideal points are called Pareto optimal solutions which are optimal values to satisfy various objectives. They are used when multiple conflicting objectives need to be optimized simultaneously. Once these Pareto optimal solutions are reached, enhancing one objective would invariably worsen others (Farsi et al., 2020). Multi-objective algorithms frequently employ Pareto selection-based techniques, where the Pareto front represents the trade-offs between conflicting objective (B. Li and Wang, 2019; Sadreddini et al., 2018). In the context of this study, the objectives are the net work output and the energy efficiency. By using the Pareto front, users can choose from a diverse set of solution between these objectives, where they may choose to maximize one and minimize the other, or choose a middle point, that satisfies both objectives partially.

A flowchart representing the whole multi-objective optimization method is given in Fig. 24.

7.4. ANN accuracy and error evaluation

The MSE is one of the most important evaluation metrics used to gauge the effectiveness of the ANN. The MSE determines the average of the squared errors between the estimated and actual values. A lower MSE value is preferred as it indicates the neural network’s overall performance.

Another key aspect of the neural network evaluation is the coefficient of determination, or the R-value. An R-value close to 1.0 signifies that the model’s predictions closely match the actual values, indicating a

strong predictive capability of the neural network. Conversely, an R-value of 0.0 suggests a random relationship between the predicted and actual values, indicating poor predictive performance by the neural network. Through a series of trial-and-error runs, the optimal number of operating parameters is determined for the Bayesian Regularization training method, ensuring that the neural network model performs at its best. The ANN model results are given in Table 12.

In Fig. 25, it is evident that the trained and test data closely match each other. Although there is an initial deviation from the best value line, they eventually converge and align. Moreover, the MSE value, as indicated in Table 12, is exceptionally low, and the R-value is equal to 1 which can be seen in Fig. 25. This outcome is highly favourable and indicative of a suitable optimization study. The close alignment of data points around the unity line in Fig. 26 signifies the successful and reliable application of the ANN.

7.5. Optimization Results

In the context of MOO using GA method, a decision-making procedure is necessary to choose the ultimate optimal solution from the Pareto front. The ANN objective function is fine-tuned and optimized by the GA to identify the most suitable solutions. The optimum values found by the Genetic Algorithm Pareto font analysis is given in the table below in Table 13.

The Pareto Front in Fig. 27 displays the optimal points for the MERS for the energy efficiency and the net work of the system. Points A, B and C represents the maximum net work, optimal performance, and maximum energy efficiency respectively, and are given below in Table 14 along with their corresponding combinations of input parameters.

From Fig. 27 and Table 14, we can notice that as the net work by the system increases, the energy efficiency of the overall cycle consistently decreases. Point A has the highest energy efficiency at 79.13 %, accompanied by a net-work output of around 4664.03 kW. Conversely, the maximum net work is achieved at point C, reaching 9018.43 kW, while the energy efficiency is 68.87 %. In multi-objective optimization, a decision-making procedure is required to select the final optimal solution from the Pareto frontier of optimal solutions. This process entails determining a hypothetical ideal point where both objective functions reach their optimal values independently. To aid in this decision-making process, TOPSIS was used to find the optimal combination between net work and energy efficiency. As there are only two objectives with equal importance, a weight of 0.5 (from a total of 1) was given to both of them. Points closer to the ideal point represent more optimal solutions. In this context, point B appears to be the closest to the ideal point, having 74.95 % efficiency and 6890.55 kW of net work for the proposed system. The efficiency is notably high due to the optimized balance achieved through the multi-objective optimization which identifies the most efficient configurations against the input parameters. In this study, we only considered limited conditions as input parameters leaving room to find more efficient values by considering more factors affecting the system.

8. Conclusion

The primary goal of this study was to maximize the efficiency of waste heat recovery from marine vessels. To achieve this, we proposed the “Marine Energy Recovery System (MERS)”, an integrated system that combines a supercritical/transcritical CO₂ cycle, a flash-tank in the refrigeration cycle, and an ORC to recover the lost heat from the gas cooler. This unique design introduces the ORC as a bottoming power cycle, as opposed to usually being the topping cycle for waste heat recovery systems. The high specific heat and low specific volume of supercritical CO₂ allow for compact turbomachinery and heat exchanger designs, meeting the demand for small, yet efficient, power and refrigeration systems in marine vessels like MERS. The parametric analysis

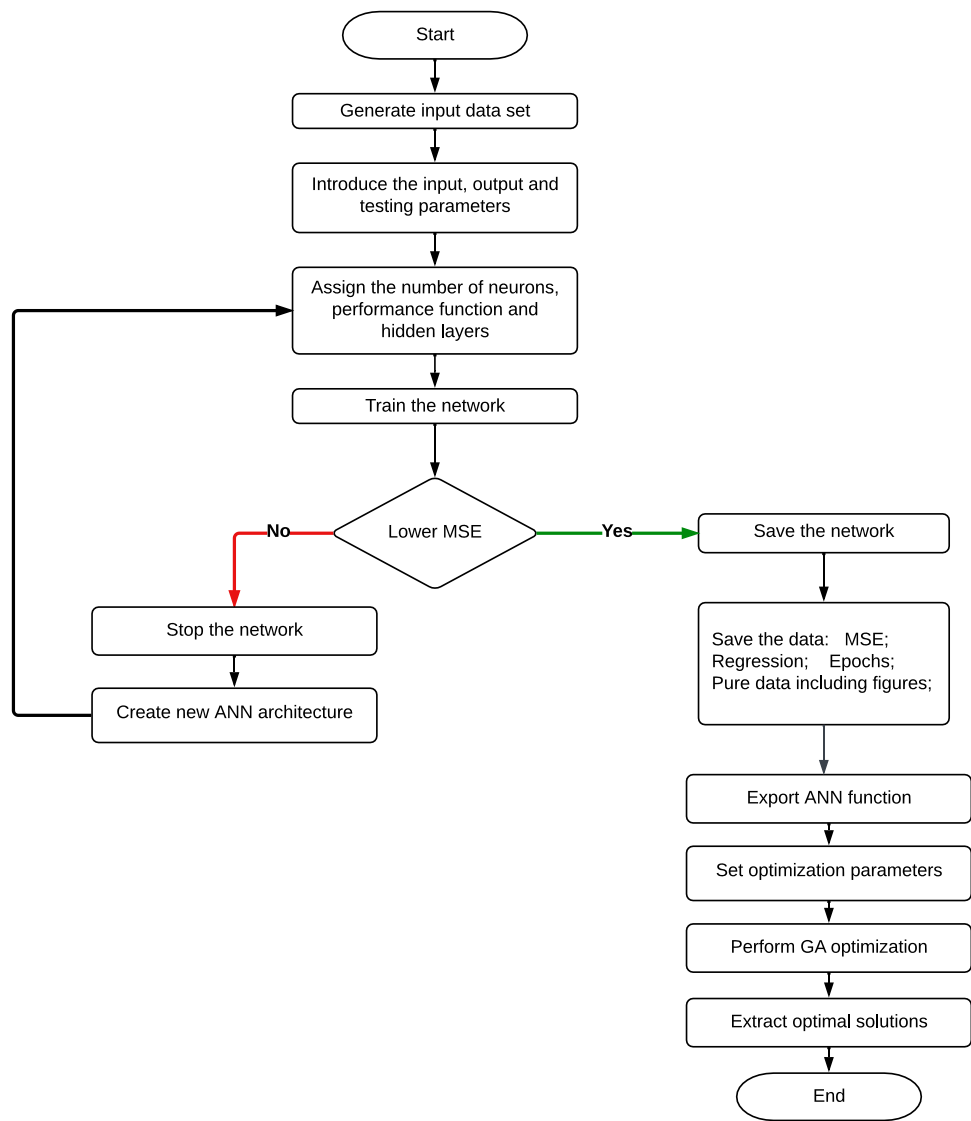


Fig. 24. Flowchart for the MERS optimization.

Table 12
ANN model results.

	Observations	MSE	R
Training	95	0.0305	1.0000
Validation	0	NaN	NaN
Test	15	0.0487	1.0000

and optimization reveal that MERS significantly enhances both energy and exergy efficiencies, with improvements of approximately 28.22 % and 39.06 %, respectively, compared to the reference system. Key findings of this study include:

- With higher gas cooler (GC) pressure, the net work of the system decreases while the cooling capacity increases. This indicates that the system becomes more efficient in its refrigeration function under higher GC pressure. Furthermore, the exergy destruction trends show that the gas cooler pressure also affects component irreversibility, with HTR and LTR showing significant increases in exergy destruction at higher pressures, meaning more energy is wasted due to inefficiencies within these parts of the system.
- Higher evaporation temperatures reduce the net work output as well as refrigeration capacity. This indicates that the system’s ability to

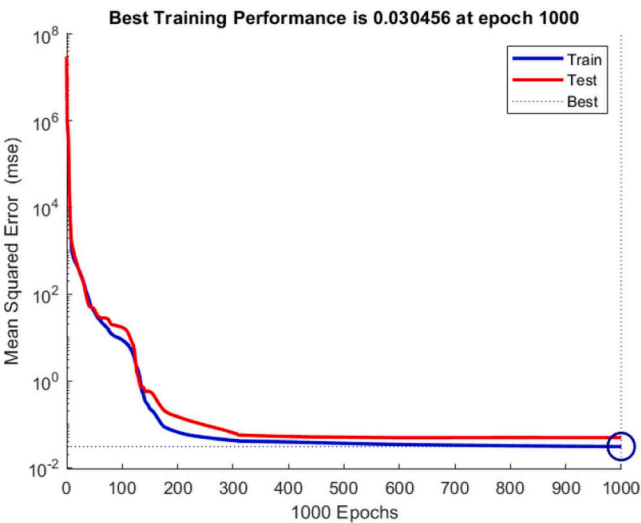


Fig. 25. Performance state analysis of ANN model.

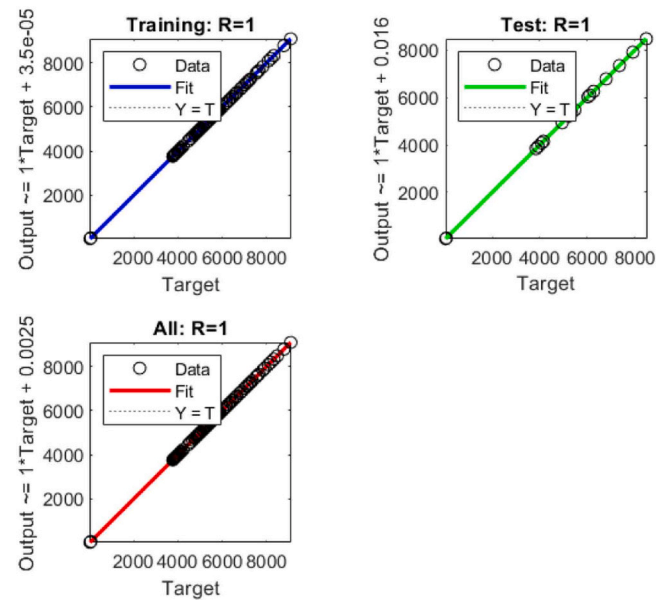


Fig. 26. Accuracy levels of test and training data as well as their verification.

Table 13
Output values found by performing Genetic Algorithm for the MERS.

Net Work output (kW)	Energy efficiency (%)	Net Work output (kW)	Energy efficiency (%)
4692.266	78.86741	4783.327	78.90686
5863.543	76.79807	6379.669	75.51998
8786.454	69.96032	8328.343	71.54679
7505.926	73.65113	8733.59	70.17196
7535.584	73.58423	8922.009	69.36324
5110.241	78.21858	8733.59	70.17196
7669.423	73.276	7959.136	72.56669
7166.472	74.38644	5457.112	77.65233
7902.35	72.71093	4695.129	79.07506
9018.428	68.87333	5853.916	76.923
6904.017	74.92398	4851.939	78.77686
4842.992	78.19272	8905.624	69.44035
8922.009	69.36324	7505.926	73.65113
6767.057	75.07854	7251.459	74.2071
8745.309	70.12591	6386.018	75.93137
4664.025	79.1347	4664.025	79.1347
7179.352	74.35944	7261.106	74.18656
7906.31	72.70096	7669.423	73.276
8905.624	69.44035	7166.472	74.38644
4796.742	78.88138	8786.454	69.96032
7952.195	72.58448	5110.744	78.29206
8272.824	71.7109	7959.136	72.56669
6678.375	75.37002	7902.35	72.71093
5157.996	77.55758	5164.78	77.84283
5156.466	78.20715	9010.844	68.91444
5461.129	76.74698	9018.428	68.87333
4847.285	78.44726	7535.584	73.58423
7699.423	73.20539	8350.062	71.48134
5900.509	75.95128	6664.466	75.39711
9010.844	68.91444	5933.172	76.77683
5481.002	77.40226	6343.486	76.01181
5919.5	76.80207	6904.017	74.92398
5919.348	76.45223	6890.551	74.95098
7699.423	73.20539	6339.963	75.93265
6765.118	75.20007	8269.694	71.72002

convert heat into useful work diminishes as the evaporation temperature rises, and the system’s cooling efficiency is adversely affected by increased evaporation temperatures.

- Turbine inlet temperature was also considered in the parametric analysis, showing a decreasing pattern in net work performance but an increasing one in energy efficiency, which is favorable for the

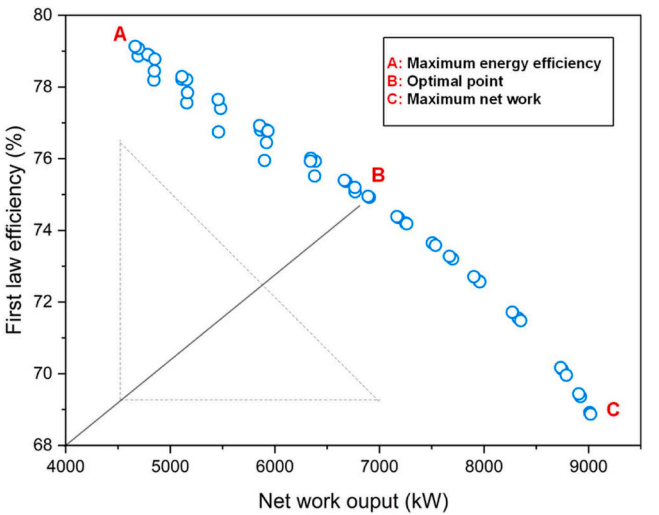


Fig. 27. Pareto front derived from results of multi-objective optimization.

system’s performance as the temperature increases. Evaporation temperature and turbine inlet temperature impact exergy destruction, with higher temperatures increasing the destruction rate.

- The ANN model from the optimization study demonstrates excellent predictive capabilities with low MSE (0.0305) and an R-value of 1, providing suitable input and target parameters for multi-objective optimization using GA, which identifies optimal operating conditions with 74.95 % energy efficiency and 6890.55 kW net work output.

Considering these behaviors, the operating conditions for the new system have been set to achieve an increased net work output of 3832.76 kW, a significant improvement from the reference system. The net-cooling capacity increases from 3082.08 kW to 3597.78 kW, a positive change for the system. The COP increases from 2.72 to 2.99, indicating better performance in MERS cooling. From the parametric and multi-objective optimization analysis, it can be concluded that MERS, with its new configuration, including the addition of the flash-tank and ORC, has improved the cooling capacity and overall performance by a significant amount, fulfilling the study’s primary objective of recovering waste heat from marine shipboards.

Future scope and recommendation

The adoption of MERS in maritime applications could lead to significant improvements in energy efficiency and environmental sustainability. Ships and marine vessels rely heavily on cooling systems, whether for refrigeration, air conditioning, or various industrial processes. Thus, applying this configuration will help to achieve better cooling capacity while keeping energy and exergy efficiency high. However, it is essential to acknowledge that certain challenges and limitations in the MERS design can be found because of the extreme temperature disparity due to heat capacity mismatch between the different streams, especially in systems that employs numerous heat exchangers. This temperature difference results in exergy losses within the components as well as a substantial portion of the total irreversibility in the system. Mitigating these losses and optimizing the system further remain areas of potential research and development.

Ethics statement

The study did not require review or approval by an ethics committee because it solely involved a software-based simulation of a combined power and cooling system. The authors developed a model, validated it

Table 14
Optimized values from Pareto front results.

Condition		Input	Temperature of the gas cooler outlet, °C	Temperature of the turbine inlet, °C	Output	Energy efficiency (%)
		Pressure in a gas cooler, MPa			Net Work (kW)	
A	Maximum energy efficiency	9.500063	50	534.59	4664.03	79.1347001
B	Optimal	9.500063	50	461.755	6890.55	74.95098
C	Maximum net work	9.500063	50	355.687	9018.43	68.8733276

against a reference model, and optimized the input parameters. Due to the non-invasive nature of the study and the lack of involvement with human subjects, informed consent was not necessary.

CRedit authorship contribution statement

Yasin Khan: Writing – review & editing, Supervision, Project administration, Conceptualization. **Mohammad Monjurul Ehsan:** Writing – review & editing, Supervision, Project administration, Conceptualization. **Fairooz Nanzeeba:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Tajwar A Baigh:** Writing – original draft, Visualization, Investigation, Formal analysis, Data curation. **Afrida Kabir:** Writing – original draft, Visualization, Investigation, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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